Operator Learning for Near Symmetric Matrices

Aleksander Markiewicz

Christ Church

University of Oxford

A thesis submitted for the degree of

M.Sc. in Mathematical Modelling and Scientific Computing

Trinity Term 2025

Acknowledgements

I would like to thank my supervisors Prof. Y. Nakatsukasa and Dr. N. Pritchard for their time and insightful remarks, not only on the mathematics, but also for writing and presenting my work. I would also like to thank Prof. N. Boullé for his correspondence.

Abstract

In this dissertation, we shall focus on the learning of near-symmetric operators. We present two methods commonly used in operator learning, namely dynamic mode decomposition (DMD) and the Nyström approximation. We shall present two notions of asymmetry, before presenting the connections and differences between them.

Thence, we focus on the Nyström approximation, by presenting a hypothesis that we can bound the error between a near-symmetric operator and its Nyström approximant by a function of the asymmetry of operator.

Lastly, we compare the effectiveness of the Nyström approximation against a neural network and consider near-symmetric cases where each method may be more advantageous.

Contents

Chapter 1

Introduction

1.1 Preliminaries

This dissertation contains results from numerical experiments. All codes pertinent to these experiments are available here. Generative AI was used in three limited ways: to help debug code, to help find references, and to check spelling and grammar.

In Chapter 1, all writing after Remark 1.7 is entirely my own, with the exception of Definition 1.8 and Lemma 1.9. Similarly in Chapter 2, all contributions are my own, with the exception of the subsection introducing neural networks 2.5.1.

1.2 Operator Learning

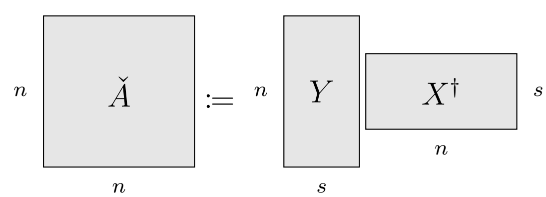
There are strong connections between the fields of operator learning and numerical linear algebra. Recently, there has been a substantial influx of interest in the capabilities of artificial intelligence (AI) and machine learning (ML) more broadly, for uses in protein structure prediction [10] to computer vision [15] and quantum physics [9].

In general, operator learning can be defined formally on some domain Ω, and functions u and v correspond to function spaces $\mathcal{U}$ and $\mathcal{V}$ respectively. Then, we may define $\mathcal{A} : \mathcal{U} \to \mathcal{V}$ as some mapping between the function spaces. In this general context, we shall denote $\mathcal{A}[u] = v$ to be the explicit relation between the functions by the operator. Let us now assume that u and $\mathcal{A}[u]$ are known and u is free to choose. The goal of our study is to choose an input or sequence of inputs $\{u_i : i \in 1, ..., n\}$, where $n \in \mathbb{N}$, in order to learn the action of the operator $\mathcal{A}$. We should note that this statement implicitly reveals the richness of the field, since we seek to find the most information about $\mathcal{A}$ in as few queries u_i as possible. One should note that in general, the field of operator learning applies to both linear and non-linear operators.

In the context of linear algebra, we may take $A \in \mathbb{C}^{n\times n}$ to be the operator we wish to learn and some input $X \in \mathbb{C}^{n\times s}$. Then, from input X and output AX, we wish to approximate matrix A. We have not yet specified size of s relative to n. Indeed, if $s = 1$, it is clear we recover the form of a standard linear system. One should also note that our goal is always to find a good approximation of A in considerably less than n queries. This is because for any matrix $A \in \mathbb{C}^{n\times n}$, we may apply the canonical vectors denoted e_i, where e_i represents the i^{th} column of the identity matrix, I_n. By querying A by inputs $\{e_1, \dots, e_n\}$, we recover the matrix in its entirety. In the following sections, we will introduce two methods for operator learning, namely dynamic mode decomposition and the Nyström approximation. For the rest of the dissertation, we shall only consider real matrices, although the work we have done can be easily generalised to complex operators.

1.2.1 Dynamic Mode Decomposition

Let us now assume that $s > 1$, and define $Y = AX$ as a matrix equation. We may naïvely take the Moore-Penrose pseudoinverse of our known input matrix X, denoted $X^\dagger$, to find that $\check{A} := YX^\dagger \simeq A$. For now, we assume that $\text{rank}(A) = n$. By inspecting the dimensions of the above expression, it is clear to see that under (as yet undefined) 'perfect' conditions, we can reconstruct a rank-s approximation of matrix A, for some $s < n$.



(1.1)

We can see this trivially by considering the column-wise definitions $Y = [y_1 \cdots y_s]$ and $X = [x_1 \cdots x_s]$, where x_i and y_i denote the i^{th} columns of X and Y respectively, for $1 \leq i \leq s$. It follows that the right-hand side of (1.1) can be thought of as s rank-1 superposed matrices. Alternatively, letting Q_1R_1 and Q_2R_2 be the thin QR-decompositions of X and Y correspondingly, we may rewrite $YX^\dagger = Q_2R_2R_1^{-1}Q_1^{\text{T}}$, where we have assumed the input X to be of full column rank. It follows that $R_2R_1^{-1} \in$

$\mathbb{C}^{s\times s}$ can be decomposed into its singular value decomposition (SVD) $R_2R_1^{-1} = PSZ^{\mathrm{T}}$, and after letting $\hat{U} := Q_2P$ and $\hat{V}^{\mathrm{T}} := Z^{\mathrm{T}}Q_1^{\mathrm{T}}$, we find that $\check{A} = \hat{U}S\hat{V}^{\mathrm{T}}$, as the thin SVD with s singular values.

However, it is important to note that the above decomposition is in general not equal to the truncated SVD of A, given by $A_s := \sum_{i=1}^{s} \sigma_i u_i v_i^{\mathrm{T}}$, where σ_i refers to the i^{th} singular value of A, and u_i and v_i to the i^{th} left and right singular vectors respectively. This is because even if $\operatorname{rank}(X) = s$, $XX^{\dagger} \neq I_n$, but instead a projector onto the column space of X, denoted $\operatorname{col}(X)$. The benefit of this is that $\check{A}$ interpolates A onto X precisely, as we have $\check{A}X = AXX^{\dagger}X = AX$ by the properties of the pseudoinverse. However we only have that, $\check{A}$ is the action of the matrix A onto the subspace spanned by X, and hence any action of A not contained in this space is lost. This presents an issue in operator learning, as the legitimacy of such an approximation depends on $\operatorname{col}(AX) \subseteq \operatorname{col}(X)$, or at least to a good approximation. The above form the motivations of the dynamic mode decomposition, and so we shall introduce it formally now.

Definition 1.1 (Dynamic Mode Decomposition)**.** Let $X, Y \in \mathbb{R}^{n\times r}$, with $Y = AX$ and let $\check{A} := YX^{\dagger}$. Let the full SVD of X, where $\operatorname{rank}(X) = q$ be given by

$$X = \begin{bmatrix} \Phi & \cdots \end{bmatrix} \begin{bmatrix} \Sigma_X & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} W^{\mathrm{T}} \\ \vdots \end{bmatrix}. \tag{1.2}$$

Then, $\Phi \in \mathbb{R}^{n\times q}$, $\Sigma_X \in \mathbb{R}^{q\times q}$, and $W \in \mathbb{R}^{r\times q}$. It follows that we can rewrite $\check{A}$ using these terms:

$$\check{A} = YW\Sigma_X^{-1}\Phi^{\mathrm{T}}. \tag{1.3}$$

Let us define $\check{A}'$ to be the compression of $\check{A}$ onto the column space of X^{T}, by writing

$$\check{A}' := \Phi^{\mathrm{T}}\check{A}\Phi. \tag{1.4}$$

We shall now use the Schur decomposition of $\check{A}'$, and write $\check{A}'\bar{V} = \bar{V}T$, where $\bar{V}$ is an orthogonal matrix and T is upper triangular. Then, the eigenvalues of the DMD are given on the major diagonal of T, with corresponding eigenvectors given by the columns of $\Psi := \Phi\bar{V}$. Together, the pair (Ψ, T) yield the *dynamic mode decomposition* of (X, Y).

We now seek to find how similar pair (Ψ, T) are to the eigen-pairs of A, the following theorem and proof is stated in Tu [18].

Theorem 1.2. *If* $\text{col}(Y) \subseteq \text{col}(X)$*, the DMD eigenvalues and eigenvectors given in Definition 1.1 are the eigenvectors and eigenvalues of operator* $\check{A}$*.*

Proof. By direct substitution of the definition of $\check{A}'$ as given in Equation (1.4) along with its Schur decomposition, we find the following relations:

$$\begin{aligned} \Phi^{\mathrm{T}}\check{A}\Phi\bar{V} &= \bar{V}T \\ \Phi\Phi^{\mathrm{T}}\check{A}\Phi\bar{V} &= \Phi\bar{V}T \\ \Phi\Phi^{\mathrm{T}}\check{A}\Psi &= \Psi T. \end{aligned} \tag{1.5}$$

By the SVD of X, we know that Φ has orthonormal columns, and hence, we may decompose Y into its components contained in $\text{col}(X)$ and those in the column space perpendicular to it, denoted $\text{col}(X^{\perp})$, also given by $\text{null}(X^{\mathrm{T}})$. Then,

$$Y = \Phi\Phi^{\mathrm{T}}Y + R', \tag{1.6}$$

where $R' \in \text{null}(\Phi^{\mathrm{T}})$, given by $R' \coloneqq (I - \Phi\Phi^{\mathrm{T}})Y$. Combining this with Equation (1.3), we find that

$$\begin{aligned} \check{A} &= (\Phi\Phi^{\mathrm{T}}Y + R')W\Sigma_X^{-1}\Phi^{\mathrm{T}} \\ \check{A} &= \Phi\Phi^{\mathrm{T}}\check{A} + R'W\Sigma_X^{-1}\Phi^{\mathrm{T}}. \end{aligned} \tag{1.7}$$

Using Equation (1.7) along with Equation (1.5), we find that

$$(\check{A} - R'W\Sigma_X^{-1}\Phi^{\mathrm{T}})\Psi = \Psi T, \tag{1.8}$$

leading to the relation

$$\check{A}\Psi - \Psi T = R'W\Sigma_X^{-1}\bar{V}. \tag{1.9}$$

In particular, in the case where $\text{col}(Y) \subseteq \text{col}(X)$, we should expect $R' = 0$, and hence the entries of pair (Ψ, T) contain the eigenvalues and eigenvectors of $\check{A}$, as well as being the eigenpairs of A restricted to $\text{col}(X)$. □

Remark 1.3. Perhaps most importantly, we should note that when $R' \neq 0$, there is no guarantee that there is close resemblance between (Ψ, T) and the true eigendecomposition of $\check{A}$, nor A.

1.2.2 The Nyström Approximation

We now introduce the Nyström method often used in low rank approximations of symmetric positive semi-definite (SPSD) matrices, particularly in machine learning problems for kernel matrices, otherwise known as Gram matrices.

Definition 1.4 (Gram matrix)**.** Consider a symmetric function $\mathcal{K} : \Omega \times \Omega \to \mathbb{R}$. Let $x_1, \ldots, x_n \in \Omega$ and let $\alpha_1, \ldots \alpha_n \in \mathbb{R}$. We say $\mathcal{K}$ is a positive definite kernel function if

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j \mathcal{K}(x_i, x_j) \geq 0 \tag{1.10}$$

is satisfied for all such values, for $n \in \mathbb{N}$. Then, for $x_1, \ldots, x_n$, we can define the Gram matrix $K \in \mathbb{R}^{n\times n}$ entry-wise by $(K)_{ij} \coloneqq \mathcal{K}(x_i, x_j)$, where $(K)_{ij}$ represents the entry on the i^{th} row and j^{th} column.

Definition 1.5 (Nyström Approximation)**.** Towards the Nyström approximation, let r be a positive integer such that $r < n$, and decompose K into the blocks

$$K = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix}, \text{ where } K_{11} \in \mathbb{R}^{r\times r} \text{ and } K_{11} \succcurlyeq 0. \tag{1.11}$$

It should be clear that K and hence K_{11} are symmetric and positive semi-definite (denoted $K \succcurlyeq 0$ here) since by direct substitution, $y^{\mathrm{T}} K y = \sum_{i=1}^{n} \sum_{j=1}^{n} y_i (K)_{ij} y_j = \sum_{i=1}^{n} \sum_{j=1}^{n} y_i y_j \mathcal{K}(x_i, x_j) \geq 0$, for any vector $y \in \mathbb{R}^n$. Furthermore, since $\mathcal{K}$ was assumed a symmetric function, $(K)_{ij} = \mathcal{K}(x_i, x_j) = \mathcal{K}(x_j, x_i) = (K)_{ji}$, and hence we find that $K = K^{\mathrm{T}}$. Often in literature, another matrix $C \in \mathbb{R}^{n\times r}$ is defined by $C \coloneqq [K_{11}\, K_{21}]^{\mathrm{T}}$, to be the first r columns of K. Then, the Nyström approximation, denoted $\tilde{K}$, is defined as:

$$\tilde{K} \coloneqq C K_{11}^{\dagger} C^{\mathrm{T}} = \begin{bmatrix} K_{11} \\ K_{21} \end{bmatrix} K_{11}^{\dagger} \begin{bmatrix} K_{11} & K_{21} \end{bmatrix}. \tag{1.12}$$

For the purposes of operator learning, we have found it more illustrative to define the Nyström approximation in terms of its inputs and outputs, $X \in \mathbb{R}^{n\times r}$ and $KX \in \mathbb{R}^{n\times r}$ respectively, and we seek to find some approximation of kernel matrix $K \in \mathbb{R}^{n\times n}$. To this end, it should be clear that $C \coloneqq KX$, and $K_{11} \coloneqq X^{\mathrm{T}} K X$, and hence we may write

$$\tilde{K} \coloneqq (KX)(X^{\mathrm{T}} K X)^{\dagger} (KX)^{\mathrm{T}}. \tag{1.13}$$

Whilst the Nyström method is designed to approximate SPSD matrices, it is capable of doing so to high precision. This is because it 'views' K twice, the first time by viewing KX, and secondly by using the Gram matrix $X^{\mathrm{T}} K X$. Indeed, by considering $(X^{\mathrm{T}} K X)_{ij} = x_i^{\mathrm{T}} K x_j$, we can also describe how x_i interacts with Kx_j, yielding more information about the geometry of K than a single viewing.

1.3 Recovery of Symmetric and Near-Symmetric Matrices

1.3.1 Matrix Recovery from Queries

In this section, we aim to introduce how a special structure in the matrix A can be exploited and how it can significantly impact our abilities to recover A in its entirety. Here, we shall therefore still regard A to be an unknown operator, but we shall assume that we have some information regarding its structure.

This is important since our research was motivated by the article 'Operator Learning without the Adjoint', by Boullé, Halikias, *et al.* [2]. In other words, we have imposed another restriction upon our goals of recovering matrix A through the use of AX and X because we do not have any information on A^{T} nor its action, $A^{\mathrm{T}}X$. A 'toy example' described in the introduction of Boullé, Halikias, *et al.* [2] is if we take $A = uv^{\mathrm{T}} \in \mathbb{R}^{n\times n}$, to be a rank-1 matrix, where $u = (u_1 \cdots u_n)^{\mathrm{T}} \in \mathbb{R}^n$ and similarly for v. Then, we can take vector $x = (x_1 \cdots x_n)^{\mathrm{T}} \in \mathbb{R}^n$ such that $v^{\mathrm{T}}x$ and $u^{\mathrm{T}}x \neq 0$, and query $x \mapsto Ax$. With access to its adjoint action, $x \mapsto A^{\mathrm{T}}x$ we can recover the full matrix in just two queries:

$$uv^{\mathrm{T}}x = \begin{bmatrix} u_1v_1 & \cdots & u_1v_n \\ \vdots & \ddots & \vdots \\ u_nv_1 & \cdots & u_nv_n \end{bmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} u_1(v_1x_1 + \cdots + v_nx_n) \\ \vdots \\ u_n(v_1x_1 + \cdots + v_nx_n) \end{pmatrix}, \tag{1.14}$$

and we shall define $\mu \coloneqq (v_1x_1 + \cdots + v_nx_n)$ for brevity. By a similar argument:

$$vu^{\mathrm{T}}x = \begin{bmatrix} v_1u_1 & \cdots & v_1u_n \\ \vdots & \ddots & \vdots \\ v_nu_1 & \cdots & v_nu_n \end{bmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} v_1(u_1x_1 + \cdots + u_nx_n) \\ \vdots \\ v_n(u_1x_1 + \cdots + u_nx_n) \end{pmatrix}, \tag{1.15}$$

and we shall denote $\nu \coloneqq (u_1x_1 + \cdots + u_nx_n)$. As a result, we can form $\mu\nu uv^{\mathrm{T}}$, and hence after dividing through by $\mu\nu$, we recover uv^{T} after just two queries. Similarly, a square Toeplitz matrix can also be uniquely determined by just two queries of itself and its adjoint. Perhaps therefore, we may expect there to be a useful reduction in the number of queries needed to recover a matrix as restrictive as a symmetric matrix?

A lemma for the recovery of symmetric matrices is provided in Halikias and Townsend [6], and we shall state it here as it shall be pivotal to our discussions.

Lemma 1.6. *A symmetric matrix $A_{sym} \in \mathbb{R}^{n\times n}$ cannot be uniquely determined in less than n queries.*

Proof. Suppose matrix A_{sym} satisfies $A_{\text{sym}} x_j = y_j$, for $1 \leq j \leq n-1$. Let us now consider matrix $B_{\text{sym}} = A_{\text{sym}} + vv^{\mathrm{T}}$, for any non-trivial vector v where $v \perp \text{span}(x_1, \ldots, x_{n-1})$. It follows that B_{sym} is symmetric since it is a symmetric matrix appended by another. Though $B_{\text{sym}} \neq A_{\text{sym}}$, we still have that $B_{\text{sym}} x_j = y_j$, for all j. □

Remark 1.7. The restrictions above have therefore pointed to the Nyström method for approximating matrices as the most useful tool for us in our endeavours. Without access to A^{T}, we can at least recover it trivially for symmetric matrices. Interesting questions arise in the cases that $A \simeq A^{\mathrm{T}}$; how well does the Nyström approximant $\tilde{A}$ approximate A if A is not symmetric? Can we bound the error $\|A - \tilde{A}\|$ in terms of its asymmetry? For matrices where $A \simeq A^{\mathrm{T}}$, we shall call them *near-symmetric* matrices, and give two definitions for how we may quantify their asymmetry.

Throughout our work, we had used Equation (1.13) for when we refer to the Nyström approximation. However, in the case of near-symmetric matrices, there is a subtle ambiguity in our choice, in particular when we impose a restriction upon the transpose. The final term in the Nyström approximation as we have presented it is an adjoint operation, since $(AX)^{\mathrm{T}} = X^{\mathrm{T}} A^{\mathrm{T}}$, for some sketching matrix X. Since our matrices are assumed to be near-symmetric, in addition to the standard expression presented in Equation (1.13), let us define $\tilde{\delta}$ to be the 'amount of asymmetry' in our matrix, though the form of $\tilde{\delta}$ for now remains undefined. Then, let us define another possible approximation, what we shall call the Nyström-$\tilde{\delta}$ approximation, denoted $\tilde{A}_{\tilde{\delta}}$, for the $\tilde{\delta}$-near-symmetric matrix A. In other words,

$$\tilde{A}_{\tilde{\delta}} \coloneqq (AX)(X^{\mathrm{T}} AX)^{\dagger} X^{\mathrm{T}} A. \tag{1.16}$$

To remove any ambiguity in our work, we shall now quantify the difference between the Nyström approximation, denoted $\tilde{A}$ here, and the Nyström-$\tilde{\delta}$ approximation in the spectral and Frobenius norms, defined by $\xi = 2, F$:

$$\|\tilde{A}_{\tilde{\delta}} - \tilde{A}\|_{\xi} = \|(AX)(X^{\mathrm{T}} AX)^{\dagger} \left[X^{\mathrm{T}} A - (AX)^{\mathrm{T}} \right] \|_{\xi} \tag{1.17}$$

$$= \|(AX)(X^{\mathrm{T}} AX)^{\dagger} \left[X^{\mathrm{T}} (A - A^{\mathrm{T}}) \right] \|_{\xi}. \tag{1.18}$$

It will occur that the skew-symmetric matrix $A - A^{\mathrm{T}}$ shall frequently emerge in our analysis, so we shall formally define it here; and denote $\tilde{\delta} := A - A^{\mathrm{T}}$. It trivially follows that

$$\|\tilde{A}_{\tilde{\delta}} - \tilde{A}\|_{\xi} = \|(AX)(X^{\mathrm{T}}AX)^{\dagger}\left[X^{\mathrm{T}}\tilde{\delta}\right]\|_{\xi}. \tag{1.19}$$

Let us now take $AX = QR$ to be the thin QR-decomposition for our sampled matrix. Then, we may write

$$\begin{aligned}\|\tilde{A}_{\tilde{\delta}} - \tilde{A}\|_{\xi} &= \|(QR)(X^{\mathrm{T}}AX)^{\dagger}\left[X^{\mathrm{T}}\tilde{\delta}\right]\|_{\xi} && (1.20)\\ &= \|R(X^{\mathrm{T}}AX)^{\dagger}\left[X^{\mathrm{T}}\tilde{\delta}\right]\|_{\xi}. && (1.21)\end{aligned}$$

Consider first the case where $\mathrm{col}(AX) \subseteq \mathrm{col}(X)$, where $X \in \mathbb{R}^{n\times s}$ with $s \leq \mathrm{rank}(A) = r$. Then kernel matrix $(X^{\mathrm{T}}AX)^{\dagger}$ is invertible and we may write $(X^{\mathrm{T}}AX)^{-1}$. Let us also assume that X is of full column rank. Then we can decompose the identity matrix as $I_r = (X^{\mathrm{T}}Q)^{-1}(X^{\mathrm{T}}Q)$. Combining this with Equation (1.21), we see that

$$\begin{aligned}\|\tilde{A}_{\tilde{\delta}} - \tilde{A}\|_{\xi} &= \|(X^{\mathrm{T}}Q)^{-1}(X^{\mathrm{T}}Q)R(X^{\mathrm{T}}AX)^{-1}\left[X^{\mathrm{T}}\tilde{\delta}\right]\|_{\xi} && (1.22)\\ &= \|(X^{\mathrm{T}}Q)^{-1}(X^{\mathrm{T}}AX)(X^{\mathrm{T}}AX)^{-1}\left[X^{\mathrm{T}}\tilde{\delta}\right]\|_{\xi} && (1.23)\\ &= \|(X^{\mathrm{T}}Q)^{-1}\left[X^{\mathrm{T}}\tilde{\delta}\right]\|_{\xi}. && (1.24)\end{aligned}$$

This is an insightful relation as it shows us the difference between the two approximations in terms of the skew-symmetric matrix which comprises matrix A, since $\tilde{\delta}$ is twice the skew-symmetric matrix, $\frac{1}{2}(A - A^{\mathrm{T}})$. Since we may assume that the eigenspectrum of our sketching matrix X is known and Q has orthonormal columns, we would only require an approximation of the magnitude of $\|X^{\mathrm{T}}\tilde{\delta}\|_{\xi}$ in order to find a good estimation of the error value. Although we shall adhere to the standard definition of the Nyström approximation throughout the rest of our work, it is useful to consider how we may append these discussions to better suit the adjoint-free cases of operator learning.

The second case we shall consider shall be for $X \in \mathbb{R}^{n\times s}$, with $s > \mathrm{rank}(A) = r$. In this case, it is clear that $(X^{\mathrm{T}}AX)$ is not invertible and so we cannot simplify any further than

$$\|\tilde{A}_{\tilde{\delta}} - \tilde{A}\|_{\xi} = \|(X^{\mathrm{T}}Q)^{\dagger}(X^{\mathrm{T}}AX)(X^{\mathrm{T}}AX)^{\dagger}\left[X^{\mathrm{T}}\tilde{\delta}\right]\|_{\xi}. \tag{1.25}$$

We may write $P_{X^{\mathrm{T}}AX} := (X^{\mathrm{T}}AX)(X^{\mathrm{T}}AX)^{\dagger}$ to be the projector onto the column space of $X^{\mathrm{T}}AX$, and hence the geometrical interpretation of the final two terms of Equation (1.25) in this case is the projection of $X^{\mathrm{T}}\tilde{\delta}$ onto the range of $X^{\mathrm{T}}AX$.

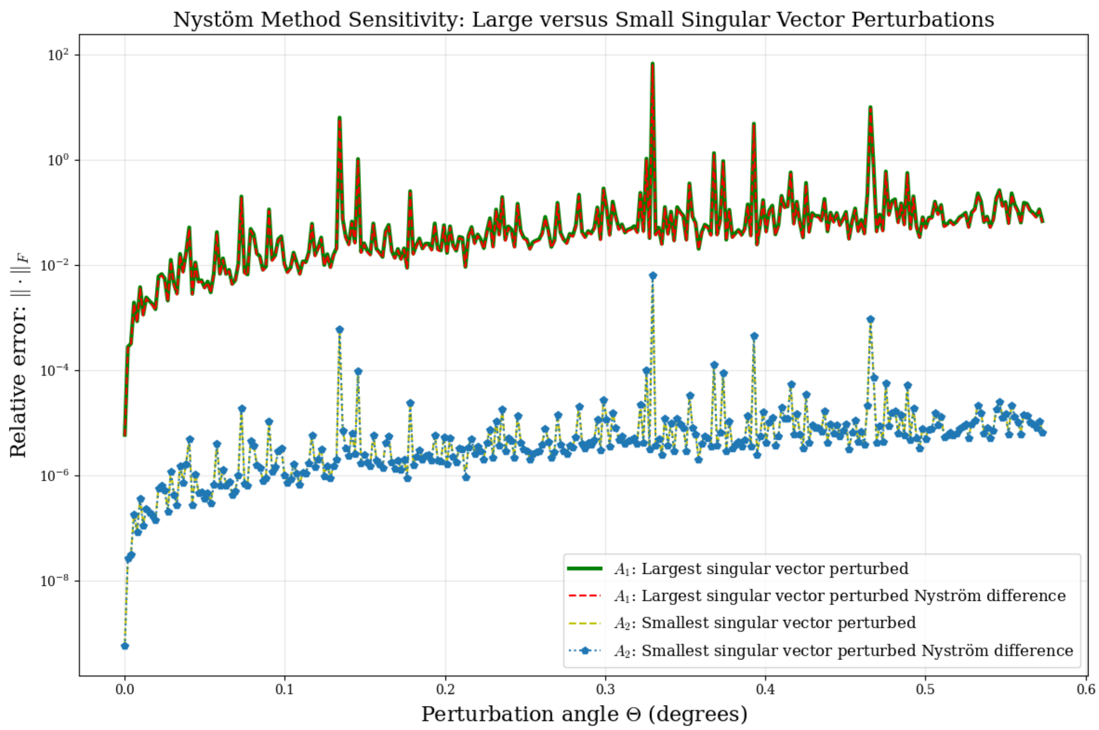


Figure 1.1: A plot showing the relative errors between the Nyström approximation and the Nyström-$\tilde{\delta}$ approximation, for one near-symmetric matrix with the right singular vector corresponding to the largest singular value perturbed and for another near-symmetric matrix with the right singular vector corresponding to the smallest singular value perturbed.

To demonstrate our findings, in Figure 1.1, we had formed two near-symmetric matrices by taking a right singular vector from the thin SVD of a symmetric matrix, and perturbing it by some perturbation angle Θ. The first matrix, A_1, is formed by perturbing the first right singular vector of symmetric matrix A_{sym}, corresponding to the largest singular value and A_2 is formed by perturbing the last right singular vector corresponding to the smallest singular value. Then, we form both Nyström and Nyström-$\tilde{\delta}$ approximations with these matrices, using the same Gaussian sketching matrix $X \in \mathbb{R}^{n\times r}$, where $r = \text{rank}(A_1) = \text{rank}(A_2)$. We then consider the relative error in the Frobenius norm of both matrices, given by the solid green and dotted green lines in Figure 1.1 respectively. Next, we compute $A_1X = Q_1R_1$ and $A_2X = Q_2R_2$ to be the thin QR-decompositions of the sketched matrices respectively, and use our expression in Equation (1.24) in order to see if our bound is a good match. These are then plotted in Figure 1.1 in a red dotted line for the analysis of A_1 and a blue dotted line with points for the analysis of A_2.

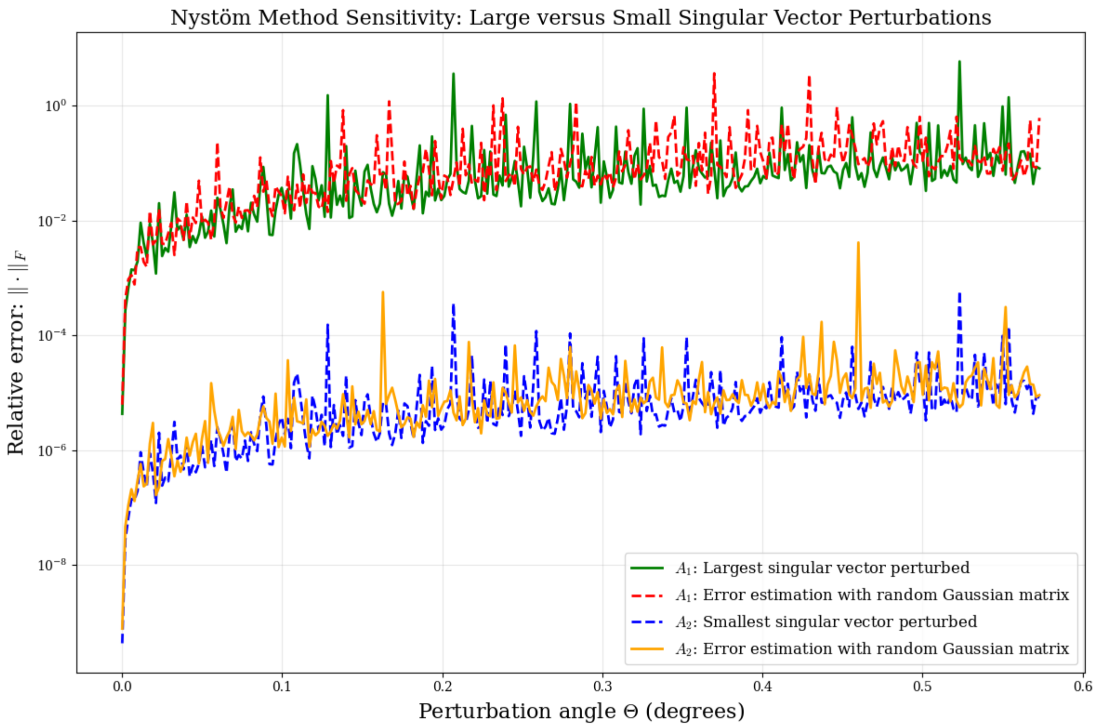


Figure 1.2: A plot showing the utility in error estimation between $\tilde{A}$ and $\tilde{A}_{\tilde{\delta}}$ in the Frobenius norm by using a random matrix instead of computing the QR-decomposition of the sketched matrices.

Perhaps the most interesting property of this relation is the use of Gaussian sketching matrices. This is because it follows that $(X^{\mathrm{T}}Q)^{-1}$ in Equation (1.24) is the inverse of another square Gaussian matrix. Indeed, without computing the QR-factorisation of the sketched matrices, and instead forming any $r \times r$ Gaussian matrix and then inverting it, we also recover a good approximation for the relative error between the Nyström approximation and the Nyström-$\tilde{\delta}$ approximation. In our case, we had taken ten Gaussian matrices, $X_1, \ldots, X_{10}$, and applied the method above in each case. We had then used the `numpy.mean()` tool from the Python library in order to reduce the effect of any numerical instabilities. The result is shown in Figure 1.2, we see a good approximation of the relative error between $\|\tilde{A} - \tilde{A}_{\tilde{\delta}}\|_F$ though only the use of Gaussian matrices along with $\tilde{\delta}$ and the sketching matrices used. The relative errors for A_1 and A_2 are given by solid green and dashed blue lines respectively, and the error approximations are given by dashed red and solid orange lines respectively.

It is apparent at this stage that the notion of near-symmetry will frequently arise in our studies. In the remainder of this chapter and the following, we shall introduce two definitions of matrix near-symmetry. We shall then show where these definitions come from, before finding similarities and differences between them through both mathematical reasoning and numerical experiments.

1.3.2 Near-Symmetric Matrices

The first definition of asymmetry of some matrix A comes from a definition in Boullé, Halikias, *et al.* [2].

Definition 1.8 (ϵ-Near-Symmetry)**.** Let $A \in \mathbb{R}^{n\times n}$, with $\operatorname{rank}(A) = r$ and let $A = U_A\Sigma_A V_A^{\mathrm{T}}$ be the thin SVD of A. We say that A is ϵ-near-symmetric if its left and right subspaces are ϵ close. In other words, there exists an orthogonal matrix $Q \in \mathbb{R}^{r\times r}$ such that

$$\|Q - U_A^{\mathrm{T}} V_A\|_2 \leq \epsilon. \tag{1.26}$$

Perhaps the most intuitive way of understanding this quantity of near-symmetry is to consider a symmetric matrix first. It follows that by the spectral theorem, $U_A = V_A$, and hence the SVD of A can be written as $A = U_A\Sigma_A U_A^{\mathrm{T}}$. Then, using (1.26), we have that $\|Q - U_A^{\mathrm{T}} U_A\| \leq \epsilon$. If we think of the above definition of as an implicit minimisation of ϵ (since it would be unhelpful to consider ϵ to be very large in the study of near-symmetric matrices), then we can choose $Q = I$, which suggests that a symmetric matrix has an ϵ-near-symmetry value of 0. We shall also make use of the following lemma also stated in Boullé, Halikias, *et al.* [2] that is helpful for determining the value of ϵ in terms of the singular values of matrices U_A and V_A.

Lemma 1.9. *Let U_A and V_A be two $n \times r$ matrices with orthonormal columns. Then, let $U_A^T V_A$ have a full SVD given by $U_A^T V_A = Q_l \Sigma Q_r^T$. Then, the matrix $Q_0 = Q_l Q_r^T$ satisfies*

$$\|V_A - U_A Q_0\|_2^2 = 2\left(\min_{Q^TQ=I} \|Q - U_A^T V_A\|_2\right) = 2\left(1 - \sigma_{min}(U_A^T V_A)\right). \tag{1.27}$$

Proof. Let the full SVD of $U_A^{\mathrm{T}} V_A$ be given by $U_A^{\mathrm{T}} V_A = Q_l \Sigma Q_r^{\mathrm{T}}$ and define $Q_0 \coloneqq Q_l Q_r^{\mathrm{T}}$. Let $d = \min_{Q^{\mathrm{T}}Q=I} \|Q - U_A^{\mathrm{T}} V_A\|_2$ denote the minimum distance between matrix $U_A^{\mathrm{T}} V_A$ and the set of orthogonal $r \times r$ matrices. Following a result from unitarily invariant norms in FAN and HOFFMAN [5], if A is a square matrix with $\|A\|_2 \leq 1$ with full SVD $A = U_A \Sigma_A V_A^{\mathrm{T}}$, we have that

$$\min_{Q^{\mathrm{T}}Q=I} \|Q - A\|_2 = 1 - \sigma_{\min}(A) \tag{1.28}$$

is achieved when $Q = U_A V_A^{\mathrm{T}}$. Applying (1.28) to $U_A^{\mathrm{T}} V_A$ and rearranging yields $\sigma_{\min}(U_A^{\mathrm{T}} V_A) = 1 - d$. Let $x \in \mathbb{R}^r$ be a vector of unit norm, and define $v = V_A x$ and $u = U_A Q_0 x$, and so $\|u\|_2 = \|v\|_2 = 1$. Furthermore, $u^{\mathrm{T}} v = x^{\mathrm{T}} Q_0^{\mathrm{T}} U_A^{\mathrm{T}} V_A x = x^{\mathrm{T}} Q_r Q_l^{\mathrm{T}} Q_l \Sigma Q_r^{\mathrm{T}} x = x^{\mathrm{T}} Q_r \Sigma Q_r^{\mathrm{T}} x \geq \sigma_{\min}(U_A^{\mathrm{T}} V_A) = 1 - d$, with equality when $x = Q_r e_r$. By considering similar triangles in the (u, v)-plane, we obtain

$$\frac{1 - u^{\mathrm{T}} v}{\|u - v\|_2} = \frac{\|u - v\|_2}{2},$$

and hence $\|V_A x - U_A Q_0 x\|_2^2 = \|u - v\|_2^2 = 2\left(1 - u^{\mathrm{T}} v\right) \leq 2d$, again with equality for $x = Q_r e_r$. Taking this supremum, we find that $\|V_A - U_A Q_0\|_2^2 = \sup_{\|x\|_2=1} \|V_A x - U_A Q_0 x\|_2^2 = 2d$. □

1.3.3 Properties of ϵ-Near-Symmetry

We seek to find the relationship between an ϵ-near symmetric matrix, and the canonical angle Θ by which a perturbation has been made to the left or right eigenvector space of the symmetric counterpart of the matrix in order to form the near-symmetric matrix.

Let us begin by defining the symmetric matrix $A_{\text{sym}} \in \mathbb{R}^{n \times n}$, and let us assume $\operatorname{rank}(A_{\text{sym}}) = r < n$, and use the spectral theorem to write down its compact eigendecomposition by $A_{\text{sym}} = U \Sigma U^{\mathrm{T}}$. Let us also denote the columns of matrix U by $[u_1 \cdots u_r]$, and our perturbed matrix V by its columns $[v_1 \cdots v_r]$. Then, for the i^{th} right singular vector v_i, we can write it in terms of u_i and a unit vector orthogonal to $\operatorname{span}(U)$, in order to maintain orthogonality and hence the validity of the form of the SVD. We shall denote this vector w, for every column i. In other words, we have that

$$v_i := u_i \cos(\theta_i) + w \sin(\theta_i), \text{ for } w \perp \operatorname{span}(U), \tag{1.29}$$

for some angle θ_i, (as defined in DAVIS and KAHAN [3]). For each column, we shall define 'column-wise near-symmetry'. This can be though of as the analogue of Equation (1.26) for vectors, such that

$$\min_{q^2=1} \|q_i - u_i^{\mathrm{T}} v_i\|_2 \leq \epsilon_i. \tag{1.30}$$

It follows that each $q_i = \pm 1$, in order to remain an analogue of a 1×1 orthogonal matrix. Then, inserting the definition of v_i as in Equation (1.29), we find that

$$\min_{q^2=1} |q_i - \cos(\theta_i)| \leq \epsilon_i, \tag{1.31}$$

for some ϵ_i, which bounds the degree of orthogonality of the vectors. Since we assume that perturbations in θ_i are small in the case of near-symmetric matrices, we shall make the approximation $\cos(\phi) \simeq 1 - \frac{1}{2}\phi^2$, which leads us to the expression

$$\epsilon_i \simeq \frac{1}{2}\theta_i^2. \tag{1.32}$$

We now seek to find how this relation can be generalised into its matrix analogue Equation (1.26). We do this by finding the relation between ϵ_i, the 'near-symmetry' value of column i, and ϵ, for each column $u_1, \ldots, u_r$.

Consider a near-symmetric matrix $A' = U\Sigma U'^{\mathrm{T}}$, where we have formed U' by perturbing the last column of U, denoted u_r. Then, as per our column-wise definition, $\|q_r - u_r^{\mathrm{T}} u_r'\|_2 = \epsilon_r$, for some $q_r = \pm 1$. By Lemma lemma 1.9 we have that $\|Q - U^{\mathrm{T}} U'\|_2 = 1 - \sigma_{\min}(U^{\mathrm{T}} U') = 1 - \cos(\theta_r)$, where the last equality is through direct calculation. It follows that $\epsilon_r \leq \epsilon$. Let us now assume there exists another column, v_s, with more perturbation such that $\theta_s > \theta_r$, and so $\cos(\theta_s) < \cos(\theta_r)$. From the definition given in Equation (1.30), and noting that for angles in $[0, \pi]$, the expression is uniquely minimised by taking $q = 1$, $\epsilon_s = 1 - \cos(\theta_s) > 1 - \cos(\theta_r) = \epsilon_r$. Therefore, in order to maintain the less than or equal to bound in Equation (1.26), we must have that

$$\max_i \epsilon_i = \epsilon. \tag{1.33}$$

Following the substitution of Equation (1.32) into Equation (1.33), we find a relation between the near-symmetry coefficient ϵ, and the angle of perturbation:

$$\epsilon = \max_i \epsilon_i \simeq \max_i \frac{1}{2}\theta_i^2.$$

As a result, we can define the canonical angle Θ as the largest angle between the left and right singular vectors of a matrix:

$$\epsilon \simeq \frac{1}{2}\Theta^2. \tag{1.34}$$

1.3.4 Ambiguities in ϵ-Near-Symmetry

A possible shortcoming of the ϵ definition for near-symmetry is apparent in Equation (1.33), since the definition does not consider the energy of the corresponding singular space. In other words, this definition is independent from the relative magnitudes of the singular values of the near-symmetric matrix.

To see this more clearly, let us consider two matrices, A_{sym}, and A', both in $\mathbb{R}^{n\times n}$, where A_{sym} is a symmetric matrix and A' is an ϵ-near-symmetric matrix that will be formed through perturbations of A_{sym}. Then, we may write the column-wise definition of the SVD of A_{sym} as:

$$A_{\text{sym}} = \sum_{i=1}^{n} \sigma_i u_i u_i^{\mathrm{T}}. \tag{1.35}$$

Let us now assume that we have formed the ϵ-near symmetric matrix A' through a perturbation of the k^{th} right singular vector from $u_k^{\mathrm{T}} \mapsto v_k^{\mathrm{T}}$. Then, by the same notation,

$$A' = \sum_{\substack{i=1\\ i\neq k}}^{n} \sigma_i u_i u_i^{\mathrm{T}} + \sigma_k u_k v_k^{\mathrm{T}}. \tag{1.36}$$

We can now represent the difference between matrices A_{sym} and A', by observing that both matrices have the same column-wise structure in all cases except when $i = k$:

$$\begin{aligned} A_{\text{sym}} - A' &= \sigma_k u_k u_k^{\mathrm{T}} - \sigma_k u_k v_k^{\mathrm{T}} \\ &= \sigma_k (u_k u_k^{\mathrm{T}} - u_k v_k^{\mathrm{T}}). \end{aligned} \tag{1.37}$$

As a result, we conclude that the ϵ-like asymmetry of the difference matrix is contained in the rank one perturbation $u_k v_k^{\mathrm{T}}$, unincorporated with σ_k.

We now demonstrate why this presents difficulties in the context of operator learning numerically, by considering two ϵ-near-symmetric matrices, A_1 and A_2, which have been formed through the perturbation of the singular vector corresponding to the largest singular value and the smallest singular value of a symmetric matrix, respectively. We will now illustrate how we form a controlled perturbation for each matrix in turn.

Consider again the symmetric matrix A_{sym}, where $A_{\text{sym}} \in \mathbb{R}^{n\times n}$ and $\text{rank}(A_{\text{sym}}) = r$. Let $A_{\text{sym}} = U\Sigma U^{\mathrm{T}}$ be the thin SVD, and let us now define w to be a unit vector orthogonal to $\text{span}(U)$, as in Equation (1.29). Using the same method as before, we shall consider the first column of U, denoted u_1, and perturb this to $u_1 \mapsto u_1\cos(\theta_1) + w\sin(\theta_1) =: v_1$. Note that the existence of such a vector w is only possible in the case that A_{sym} is not full rank and hence does not span its entire domain, as the rank-nullity theorem gives us that $w \in$ null, and so $\dim(\text{null}(A_{\text{sym}})) = n - \text{rank}(A_{\text{sym}}) = n - r$.

Then, we shall define matrix A_1 to be the matrix formed by the thin SVD $A_1 = U\Sigma V_1^{\mathrm{T}}$, where $V_1 = [v_1\, u_2 \cdots u_n]$ when written column-wise. To form A_2, we shall employ the same method, though now we shall perturb $u_r \mapsto u_r \cos(\theta_r) + w \sin(\theta_r) =: v_r$. Then $A_2 = U\Sigma V_2^{\mathrm{T}}$, with $V_2 = [u_1 \cdots u_{r-1}\, v_r]$ when written column-wise. By ensuring that $\theta_1 = \theta_r$, we can be certain that both matrices are of the same ϵ-near-symmetry.

In the following analysis, we shall denote the Nyström approximation of A_1 as $\tilde{A}_1$, given by $\tilde{A}_1 = (A_1X)(X^{\mathrm{T}}A_1X)^{\dagger}(A_1X)^{\mathrm{T}}$, and similarly for A_2 with its Nyström approximation defined by $\tilde{A}_2 = (A_2X)(X^{\mathrm{T}}A_2X)^{\dagger}(A_2X)^{\mathrm{T}}$. In this case, we shall take $X \in \mathbb{R}^{n\times s}$ to be a Gaussian matrix, with $n > s \geq r$ to be the sketch size with an oversampling of $n - s$.

We shall use the relative error estimate $\frac{\|\tilde{A}_\beta - A_\beta\|}{\|A_\beta\|}$ in the Frobenius norm, where we have written $\beta = \{1, 2\}$ for brevity. Figure 1.3 shows our results. We had used $n = 100$, and $r = 10$, the rank of the matrices. We had defined the perturbation angle to range from $\Theta = 1\times 10^{-6}$–$\frac{\pi}{2}$ rad, and taken Σ to be r singular values where $\sigma_1 = 1$ and $\sigma_r = 10^{-4}$, with the distribution being exponentially decreasing through the use of the `numpy.logspace()` functionality supplied in the NumPy Python library: Harris, Millman, *et al.* [7]. We have taken such singular values in order to demonstrate the clear differences in the accuracy of the reconstructions, whilst also ensuring that our matrices are not so poorly conditioned that the `numpy.linalg.pinv()` software cannot adequately reconstruct the pseudoinverse kernel matrix $(X^{\mathrm{T}}AX)^{\dagger}$ needed for the Nyström approximation. Taking $\kappa(A_{\text{sym}}) := \frac{\sigma_{max}}{\sigma_{\min}} = 10^4$ provides a good middle-ground for these experiments.

To further ensure that the sketching matrices X used for approximating A_1 and A_2 were suitable, we had generated a set of Gaussian matrices $\{X_i : 1 \leq i \leq \ell\}$, and repeated the experiment for each X_i. Then, we would use either `numpy.average()` or `numpy.median()` to reduce or neglect any large discrepancies, although due to the dimensions of each X_i this was overwhelmingly unlikely.

We have taken a large range of angles to perturb v_1 and v_r by in order to provide more evidence for the energy of each singular subspace to be intrinsically linked to the asymmetry of the matrix. In particular, we see that for $\Theta = \frac{\pi}{2}$, the relative error of both matrices $\tilde{A}_1$ and $\tilde{A}_2$ is bounded above by the magnitudes of the singular values corresponding to their singular vector perturbations, 1 and 1×10^{-4} respectively. In our

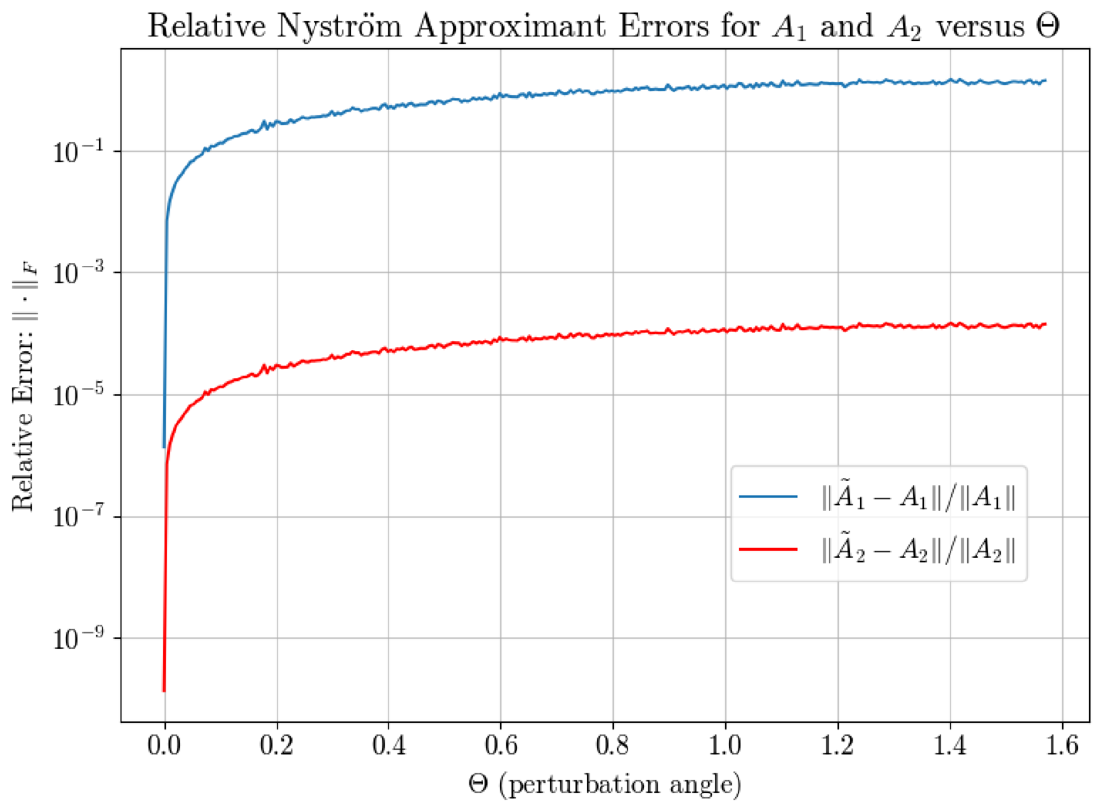


Figure 1.3: The relative errors between two Nyström approximations of ϵ-near symmetric matrices A_1 and A_2 with sketch size $s = 2r$, where $r = \text{rank}(A_1) = \text{rank}(A_2)$. The code which plots the above graph is available under `dissertation_code_9.ipynb` in the GitHub repository.

experiments, we had also considered how much of an impact the Nyström method itself had had on the approximation between the matrix and its symmetric component. In this case, we had also defined the $\tilde{A}_{\text{sym}}$ to be the Nyström approximation of A_{sym}, again using the form of Equation (1.13). We find that the relative error $\frac{\|A_{\text{sym}} - \tilde{A}_{\text{sym}}\|_F}{\|\tilde{A}_{\text{sym}}\|_F}$ was of $\mathcal{O}(10^{-13})$ with very high probability, demonstrating that the Nyström approximation is a good method to use in our case. Since this error is negligible compared to the size of the relative errors in the case of the ϵ-near-symmetric matrices, we shall claim that $\tilde{A}_{\text{sym}} \simeq A_{\text{sym}}$. Then, by substituting into Equation (1.37) we see that the difference between these matrices is approximately bounded from above by σ_k, the singular value corresponding to the perturbed singular vector. In other words, by the triangle inequality, we have that

$$\frac{\|\tilde{A}_{\mathrm{sym}} - A'\|}{\|A_{\mathrm{sym}}\|} = \frac{\|\tilde{A}_{\mathrm{sym}} - A_{\mathrm{sym}}\|}{\|A_{\mathrm{sym}}\|} + \frac{\|A_{\mathrm{sym}} - A'\|}{\|A_{\mathrm{sym}}\|} \tag{1.38}$$

$$\simeq \frac{\|A_{\mathrm{sym}} - A'\|}{\|A_{\mathrm{sym}}\|} \tag{1.39}$$

$$= \sigma_k \frac{\|u_k u_k^{\mathrm{T}} - u_k v_k^{\mathrm{T}}\|}{\|A_{\mathrm{sym}}\|}. \tag{1.40}$$

Evaluating Equation (1.40) for $\Theta = \frac{\pi}{2}$, it is clear to see that the error between the Nyström approximation of a symmetric matrix and an ϵ-near-symmetric matrix is σ_k.

Chapter 2

Near-Symmetric Matrices and the Nyström Approximation

2.1 $\tilde{\epsilon}$-Near-Symmetry

In this section, we shall now introduce some of the work we have undertaken regarding the Nyström approximation and near-symmetric matrices. To do this, we shall introduce another definition of near-symmetry, represented by a matrix rather than a scalar value. We can compare this notion of near-symmetry with our previous notion of ϵ-near-symmetry by taking the norm of this matrix.

Definition 2.1 ($\tilde{\epsilon}$-Near-Symmetric Matrices)**.** Let A be a $\tilde{\epsilon}$-near-symmetric matrix. Then, we can define $\tilde{\epsilon}$ as:

$$\tilde{\epsilon} \coloneqq A - A^{\mathrm{T}}. \tag{2.1}$$

In other words, $\tilde{\epsilon}$ is twice the skew-symmetric counterpart of the symmetric matrix that comprises A. We may write:

$$A = \frac{1}{2}\left(A + A^{\mathrm{T}}\right) + \frac{1}{2}\left(\tilde{\epsilon}\right).$$

We see that this definition of near-symmetry was motivated by $\tilde{\delta}$ as we had defined in Equation (1.18), as a result of the ambiguity of the Nyström approximation when applied to an asymmetric matrix. To introduce this section, we will motivate our studies by considering an ideal environment where we would expect the Nyström approximation to produce the exact matrix it is approximating, that is to say $\tilde{A} = A$, assuming A is symmetric, and $\tilde{A}$ represents the Nyström approximation. We show a relationship between this approximation and $\tilde{\epsilon}$, for a matrix of $\tilde{\epsilon}$-near-symmetry. The following lemma is purely motivational, as its use of a square invertible sketching matrix X is far from the standard application of the Nyström approximation. Indeed, the result is trivial by setting $X = I_n$.

Lemma 2.2. *Let $A \in \mathbb{R}^{n\times n}$ be an $\tilde{\epsilon}$-near-symmetric matrix, that is to say $\tilde{\epsilon} = A - A^T$. Let sketching matrix $X \in \mathbb{R}^{n\times n}$ be such that* $\operatorname{rank}(A) = \operatorname{rank}(X) = n$ *and* $\operatorname{col}(A) \subseteq \operatorname{col}(X)$. *Denote the Nyström approximation of A by $\tilde{A}$. Then the error $\|A - \tilde{A}\|_\xi$, for $\xi = 2, F$ is given by*

$$\|A - \tilde{A}\|_\xi = \|\tilde{\epsilon}\|_\xi.$$

Proof. Let $\tilde{A}$ represent the Nyström approximation, given by $\tilde{A} = (AX)(X^{\mathrm{T}}AX)^{-1}(AX)^T$. We shall consider matrix $AX - \tilde{A}X$, then it is clear that

$$AX - \tilde{A}X = AX - (AX)(X^{\mathrm{T}}AX)^{-1}(AX)^T X.$$

For matrix $(AX)^{\mathrm{T}}$, we shall write $(AX)^{\mathrm{T}} = X^{\mathrm{T}}(A - \tilde{\epsilon})$. It follows that

$$\begin{aligned} AX - \tilde{A}X &= AX - (AX)(X^{\mathrm{T}}AX)^{-1}X^{\mathrm{T}}(A - \tilde{\epsilon})X \\ &= AX - (AX)(X^{\mathrm{T}}AX)^{-1}X^{\mathrm{T}}AX + (AX)(X^{\mathrm{T}}AX)^{-1}X^{\mathrm{T}}\tilde{\epsilon}X \\ &= (AX)(X^{\mathrm{T}}AX)^{-1}(X^{\mathrm{T}}\tilde{\epsilon}X). \end{aligned} \tag{2.2}$$

Since A and X are full rank, we can write $(X^{\mathrm{T}}AX)^{-1} = X^{-1}(AX)^{-1}$. Multiplying both sides of (2.2) by X^{-1}, we find that

$$A - \tilde{A} = (AX)(AX)^{-1}X^{-\mathrm{T}}X^{\mathrm{T}}\tilde{\epsilon}XX^{-1},$$

and hence $\|A - \tilde{A}\|_\xi = \|\tilde{\epsilon}\|_\xi$. This concludes the proof. □

Although the above lemma is restrictive in its assumptions, it does show a correlation between the Nyström approximation and our definition of asymmetry. Such a relation may allow us to speculate that for broader categories of A and X, one may be able to derive a more general relation $\|A-\tilde{A}\| \le f(\|\tilde{\epsilon}\|)$, where f is some undetermined function.

In the following, we shall now try and derive what f shall be for more general (and hence more practical) uses. Before we begin, it should be clear that from our discussions there are two main factors that impact the magnitude of the bound $\|\tilde{A} - A\|$:

- The form and magnitude of asymmetry, $\tilde{\epsilon}$, in matrix A;
- The properties of sample matrix X, such that $\operatorname{col}(A) \subsetneq \operatorname{col}(X)$ to ensure a good operator recovery from the Nyström approximation.

To this end, we shall assume that A is an $\tilde{\epsilon}$-near-symmetric matrix, which we will take to mean $\|\tilde{\epsilon}\| \ll \|A\|$ in both the spectral and Frobenius norms. We shall assume X to be a Gaussian matrix which is not square, and hence not invertible. Finally we shall also assume that A is a rank deficient matrix, and hence also not invertible.

Theorem 2.3. *Let $A \in \mathbb{R}^{n\times n}$ be an $\tilde{\epsilon}$-near-symmetric matrix, such that $\tilde{\epsilon} = A - A^{\mathrm{T}}$ and $\|\tilde{\epsilon}\|_\xi \ll \|A\|_\xi$, where $\xi = 2, F$. Let $X \in \mathbb{R}^{n\times r}$ be a Gaussian matrix, where $r \le \operatorname{rank}(A)$. Denote $\tilde{A}$ to be the Nyström approximation of A, defined by $\tilde{A} = (AX)(X^{\mathrm{T}}AX)^{\dagger}(AX)^{\mathrm{T}}$. Then, with high probability, we can bound $\|A - \tilde{A}\|_2$ from above by*

$$\|A - \tilde{A}\|_2 \lesssim \|A_{\text{sym}} - \tilde{A}_{\text{sym}}\|_2 + \frac{1}{2}\sqrt{r}\left(2 + \sqrt{r}\|X\|_2\right)\|\tilde{\epsilon}X\|_2 + \frac{1}{2}\|\tilde{\epsilon}\|_2, \tag{2.3}$$

and we can bound $\|A - \tilde{A}\|_F$ from above with high probability by

$$\|A - \tilde{A}\|_F \lesssim \|A_{\text{sym}} - \tilde{A}_{\text{sym}}\|_F + \frac{1}{2}r\left(1 + \sqrt{r} + r\|X\|_F\right)\|\tilde{\epsilon}X\|_F + \frac{1}{2}\|\tilde{\epsilon}\|_F, \tag{2.4}$$

where A_{sym} represents the symmetric component of A, and $\tilde{A}_{\text{sym}}$ denotes the Nyström approximation of A_{sym}.

Proof. Let us begin by using the triangle inequality to decompose $\|A - \tilde{A}\|_\xi$ into more manageable and known expressions in terms of its symmetric component A_{sym}, and antisymmetric component $\frac{1}{2}\tilde{\epsilon}$. We have that

$$\|A - \tilde{A}\|_\xi \le \|A_{\text{sym}} - \tilde{A}\|_\xi + \|A - A_{\text{sym}}\|_\xi, \tag{2.5}$$

and it clearly follows that $\|A - A_{\text{sym}}\|_\xi = \frac{1}{2}\|\tilde{\epsilon}\|_\xi$. Next, we shall decompose $\|A_{\text{sym}} - \tilde{A}\|_\xi$ further and write

$$\|A_{\text{sym}} - \tilde{A}\|_\xi \le \|A_{\text{sym}} - \tilde{A}_{\text{sym}}\|_\xi + \|\tilde{A}_{\text{sym}} - \tilde{A}\|_\xi. \tag{2.6}$$

There is a large amount of literature on the Nyström method for matrix learning, for symmetric matrices A_{sym} that the method is traditionally used (see Drineas and Mahoney [4]). As a result, we shall refer to this for a suitable upper bound of

$\|A_{\text{sym}} - \tilde{A}_{\text{sym}}\|_\xi$ and instead focus on the term $\|\tilde{A}_{\text{sym}} - \tilde{A}\|_\xi$. Using our expression for the Nyström approximation, we have that

$$\|\tilde{A}_{\text{sym}} - \tilde{A}\|_\xi = \|(A_{\text{sym}}X)(X^{\mathrm{T}}A_{\text{sym}}X)^{\dagger}(A_{\text{sym}}X)^{\mathrm{T}} - (AX)(X^{\mathrm{T}}AX)^{\dagger}(AX)^{\mathrm{T}}\|_\xi. \tag{2.7}$$

Writing A in terms of A_{sym} and $\tilde{\epsilon}$, we can write

$$\tilde{A} = \left(\left(A_{\text{sym}} + \frac{1}{2}\tilde{\epsilon}\right)X\right)\left(X^{\mathrm{T}}A_{\text{sym}}X + \frac{1}{2}X^{\mathrm{T}}\epsilon\tilde{X}\right)^{\dagger}\left(\left(A_{\text{sym}} + \frac{1}{2}\tilde{\epsilon}\right)X\right)^{\mathrm{T}}. \tag{2.8}$$

We have taken the case where the kernel of the above expression is full rank; we may instead say that the pseudoinverse is a standard inverse. As a result, we can employ the Neumann series of the approximation of the inverse of a matrix by writing

$$\begin{gathered}\left(I + \left(X^{\mathrm{T}}A_{\text{sym}}X\right)^{-1}\left(\frac{1}{2}X^{\mathrm{T}}\tilde{\epsilon}X\right)\right)\left(X^{\mathrm{T}}A_{\text{sym}}X\right)^{-1} = \\ \left[\sum_{k=0}^{\infty}\left(-\left(X^{\mathrm{T}}A_{\text{sym}}X\right)^{-1}\left(\frac{1}{2}X^{\mathrm{T}}\tilde{\epsilon}X\right)\right)^{k}\right]\left(X^{\mathrm{T}}A_{\text{sym}}X\right)^{-1},\end{gathered} \tag{2.9}$$

for all $\|\left(X^{\mathrm{T}}A_{\text{sym}}X\right)^{-1}\left(\frac{1}{2}X^{\mathrm{T}}\tilde{\epsilon}X\right)\|_2 < 1$. We shall approximate this to first order; for this to be a good approximation, we shall therefore need $\|\frac{1}{2}X^{\mathrm{T}}\tilde{\epsilon}X\|_2 \ll \frac{1}{\|X^{\mathrm{T}}A_{\text{sym}}X\|_2} = \sigma_{\min}(X^{\mathrm{T}}A_{\text{sym}}X)$, where $\sigma_{\min}$ represents the smallest non-zero singular value. As a result, we require that $\frac{1}{2}\|X\|_2^2\|\tilde{\epsilon}\|_2 \ll \sigma_{\min}^2(X)\sigma_{\min}(A_{\text{sym}})$ and hence $\|\tilde{\epsilon}\|_2 \ll \frac{2\sigma_{\min}(A_{\text{sym}})}{\kappa^2(X)}$. To proceed, let us assume that X is well-conditioned and so $\kappa(X) \sim \mathcal{O}(1)$, in other words, that all entries of A are much greater in magnitude than those entries of $\tilde{\epsilon}$, for such an $\tilde{\epsilon}$-near-symmetric matrix. Therefore, we shall approximate Equation (2.9) as

$$\begin{aligned}\left(X^{\mathrm{T}}A_{\text{sym}}X + \frac{1}{2}X^{\mathrm{T}}\epsilon\tilde{X}\right)^{-1} \simeq & \left(X^{\mathrm{T}}A_{\text{sym}}X\right)^{-1} \\ & - \left(X^{\mathrm{T}}A_{\text{sym}}X\right)^{-1}\left(\frac{1}{2}X^{\mathrm{T}}\tilde{\epsilon}X\right)\left(X^{\mathrm{T}}A_{\text{sym}}X\right)^{-1}.\end{aligned} \tag{2.10}$$

Next, we shall substitute Equation (2.10) into Equation (2.8), and hence

$$\begin{aligned}
&\left(\left(A_{\text{sym}} + \frac{1}{2}\tilde{\epsilon}\right) X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X + \frac{1}{2} X^{\mathrm{T}} \epsilon \tilde{X}\right)^{\dagger} \left(\left(A_{\text{sym}} + \frac{1}{2}\tilde{\epsilon}\right) X\right)^{\mathrm{T}} \\
&\quad \simeq (A_{\text{sym}} X) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} (A_{\text{sym}} X)^{\mathrm{T}} + (A_{\text{sym}} X) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2}\tilde{\epsilon} X\right)^{\mathrm{T}} \\
&\quad + \left(\frac{1}{2}\tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} (A_{\text{sym}} X)^{\mathrm{T}} + \left(\frac{1}{2}\tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2}\tilde{\epsilon} X\right)^{\mathrm{T}} \\
&\quad + (A_{\text{sym}} X) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2} X^{\mathrm{T}} \tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} (A_{\text{sym}} X)^{\mathrm{T}} \\
&\quad + \left(\frac{1}{2}\tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2} X^{\mathrm{T}} \tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} (A_{\text{sym}} X)^{\mathrm{T}} \\
&\quad + (A_{\text{sym}} X) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2} X^{\mathrm{T}} \tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2}\tilde{\epsilon} X\right)^{\mathrm{T}} \\
&\quad + \left(\frac{1}{2}\tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2} X^{\mathrm{T}} \tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2}\tilde{\epsilon} X\right)^{\mathrm{T}}. \qquad (2.11)
\end{aligned}$$

It should be clear that the first term in Equation (2.11) is just $\tilde{A}_{\text{sym}}$. Now, once again assuming that $\|\tilde{\epsilon}\|_{\xi}$ is small for an $\tilde{\epsilon}$-near-symmetric matrix, we shall neglect any term of size $\mathcal{O}(\|\tilde{\epsilon}\|_{\xi}^2)$ or smaller. As a result, combining this with Equation (2.7), we find that

$$\begin{aligned}
\|\tilde{A}_{\text{sym}} - \tilde{A}\|_{\xi} \simeq \| &(A_{\text{sym}} X) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2}\tilde{\epsilon} X\right)^{\mathrm{T}} + \left(\frac{1}{2}\tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} (A_{\text{sym}} X)^{\mathrm{T}} \\
&- (A_{\text{sym}} X) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} \left(\frac{1}{2} X^{\mathrm{T}} \tilde{\epsilon} X\right) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1} (A_{\text{sym}} X)^{\mathrm{T}} \|_{\xi}. \qquad (2.12)
\end{aligned}$$

For brevity, let us define $C \coloneqq (A_{\text{sym}} X) \left(X^{\mathrm{T}} A_{\text{sym}} X\right)^{-1}$ and $D \coloneqq C \left(\frac{1}{2}\tilde{\epsilon} X\right)^{\mathrm{T}}$. Then, since $\tilde{\epsilon}^{\mathrm{T}} = -\tilde{\epsilon}$, we have that

$$\begin{aligned}
\|\tilde{A}_{\text{sym}} - \tilde{A}\|_{\xi} &\simeq \|D + D^{\mathrm{T}} + DXC^{\mathrm{T}}\|_{\xi} && (2.13) \\
&\lesssim \|D\|_{\xi} \|I + XC^{\mathrm{T}}\|_{\xi} + \|D^{\mathrm{T}}\|_{\xi}, && (2.14)
\end{aligned}$$

where we have used the Cauchy-Schwarz inequality in the second line. Since the spectral and Frobenius norms are unitarily invariant, we have that $\|D\|_{\xi} = \|D^{\mathrm{T}}\|_{\xi}$. We shall now move to bound the magnitude of D. Let $A_{\text{sym}} X = Q_s R_s$ be a thin

QR-decomposition, such that $Q_s \in \mathbb{R}^{n\times r}$, and $R_s \in \mathbb{R}^{r\times r}$. Then, we can write

$$\|D\|_\xi = \|C\left(\frac{1}{2}\tilde{\epsilon}X\right)^{\mathrm{T}}\|_\xi \tag{2.15}$$

$$= \|\left(A_{\mathrm{sym}}X\right)\left(X^{\mathrm{T}}A_{\mathrm{sym}}X\right)^{-1}\left(\frac{1}{2}\tilde{\epsilon}X\right)^{\mathrm{T}}\|_\xi \tag{2.16}$$

$$= \|(Q_sR_s)\left(X^{\mathrm{T}}A_{\mathrm{sym}}X\right)^{-1}\left(\frac{1}{2}\tilde{\epsilon}X\right)^{\mathrm{T}}\|_\xi \tag{2.17}$$

$$= \|R_s\left(X^{\mathrm{T}}A_{\mathrm{sym}}X\right)^{-1}\left(\frac{1}{2}\tilde{\epsilon}X\right)^{\mathrm{T}}\|_\xi \tag{2.18}$$

$$= \|\left(X^{\mathrm{T}}Q_s\right)^{-1}\left(X^{\mathrm{T}}Q_s\right)R_s\left(X^{\mathrm{T}}A_{\mathrm{sym}}X\right)^{-1}\left(\frac{1}{2}\tilde{\epsilon}X\right)^{\mathrm{T}}\|_\xi. \tag{2.19}$$

We can do this since $X^{\mathrm{T}}Q_s$ is a Gaussian $r\times r$ matrix and hence $\mathbb{P}\left(\mathrm{rank}(X^{\mathrm{T}}Q_s) = r\right) = 1$. Continuing to assume that $X^{\mathrm{T}}A_{\mathrm{sym}}X$ is invertible, we find that

$$\|D\|_\xi = \|\left(X^{\mathrm{T}}Q_s\right)^{-1}\left(X^{\mathrm{T}}A_{\mathrm{sym}}X\right)\left(X^{\mathrm{T}}A_{\mathrm{sym}}X\right)^{-1}\left(\frac{1}{2}\tilde{\epsilon}X\right)^{\mathrm{T}}\|_\xi \tag{2.20}$$

$$= \|\left(X^{\mathrm{T}}Q_s\right)^{-1}\left(\frac{1}{2}\tilde{\epsilon}X\right)^{\mathrm{T}}\|_\xi. \tag{2.21}$$

Then, from Rudelson [14] and Szarek [17], we have that, with high probability, the spectral norm magnitude of the inversion of a square Gaussian matrix can be approximated by its dimension. In our case, we have that $\|\left(X^{\mathrm{T}}Q_s\right)^{-1}\|_2 \lesssim \sqrt{r}$. As a result, we conclude that $\|D\|_2 \lesssim \frac{1}{2}\sqrt{r}\|\tilde{\epsilon}X\|_2$. Then, to bound $\|I + XC^{\mathrm{T}}\|_2$, we shall use the triangle inequality along with the same analysis of C as before, such that

$$\|I + CX^{\mathrm{T}}\|_2 \leq 1 + \sqrt{r}\|X\|_2. \tag{2.22}$$

Combining the above with Equation (2.14), we find that

$$\|\tilde{A}_{\mathrm{sym}} - \tilde{A}\|_2 \lesssim \frac{1}{2}\sqrt{r}\left(2 + \sqrt{r}\|X\|_2\right)\|\tilde{\epsilon}X\|_2. \tag{2.23}$$

Putting everything together, we have for the spectral norm that

$$\|A - \tilde{A}\|_2 \lesssim \|A_{\mathrm{sym}} - \tilde{A}_{\mathrm{sym}}\|_2 + \frac{1}{2}\sqrt{r}\left(2 + \sqrt{r}\|X\|_2\right)\|\tilde{\epsilon}X\|_2 + \frac{1}{2}\|\tilde{\epsilon}\|_2. \tag{2.24}$$

For the Frobenius norm, we cannot use the same result for inverted square Gaussian matrices. As a result, we shall use the standard inequality between the spectral and Frobenius norms, namely that for some matrix $\mathcal{M} \in \mathbb{R}^{m\times n}$, we have that $\|\mathcal{M}\|_F \leq$

$\sqrt{\mathrm{rank}(\mathcal{M})}\|\mathcal{M}\|_2$. In our case, this yields $\|(X^{\mathrm{T}}Q_s)^{-1}\|_F \lesssim r$. Further, we have that $\|I + CX^{\mathrm{T}}\| \leq \sqrt{r} + r\|X\|_F$, and so $\|D\|_F\|I + XC^{\mathrm{T}}\|_F + \|D^{\mathrm{T}}\|_F \lesssim \frac{1}{2}r\|\tilde{\epsilon}X\|_F(\sqrt{r} + r\|X\|_F) + \frac{1}{2}r\|\tilde{\epsilon}X\|_F$. We conclude that

$$\|\tilde{A}_{\mathrm{sym}} - \tilde{A}\|_F \lesssim \frac{1}{2}r\left(1 + \sqrt{r} + r\|X\|_F\right)\|\tilde{\epsilon}X\|_F. \tag{2.25}$$

Similarly, it follows that

$$\|A - \tilde{A}\|_F \lesssim \|A_{\mathrm{sym}} - \tilde{A}_{\mathrm{sym}}\|_F + \frac{1}{2}r\left(1 + \sqrt{r} + r\|X\|_F\right)\|\tilde{\epsilon}X\|_F + \frac{1}{2}\|\tilde{\epsilon}\|_F \tag{2.26}$$

and this concludes the proof. □

2.2 Numerical Experiments with the Nyström Approximation

We shall now provide some numerical results from this investigation. We had taken $n = 100$ and $r = 10$ to be the size and rank of A_{sym}, a symmetric matrix, respectively. We would then take the thin SVD of A_{sym} to be $U_A\Sigma_A U_A^{\mathrm{T}}$. To form a controlled asymmetry, we would take the right singular vector, denoted u_r, corresponding to the smallest singular value of A_{sym}, and perturb it by some small perturbation angle Θ, towards a vector $w \perp \mathrm{span}(U_A)$, with $\|w\| = 1$ to maintain orthonormality. We then denote a new matrix V_A, and form the $\tilde{\epsilon}$-near-symmetric matrix $A = U_A\Sigma_A V_A^{\mathrm{T}}$, where the complementary $\tilde{\epsilon}$ is given by matrix $A - A^{\mathrm{T}}$. This perturbation angle was increased from 0–0.1 rad to increase the asymmetry. At every angle Θ, we would form a Gaussian matrix $X \in \mathbb{R}^{n\times r}$ from the `numpy.random.randn()` Python library, before computing the Nyström approximation $\tilde{A}$, and the Nyström approximation of the symmetric matrix, $\tilde{A}_{\mathrm{sym}}$. We would then plot both sides of Equation (2.24) against perturbation angle Θ as shown in Figure 2.1.

We now seek to analyse our numerical results. The algorithm outlined above was performed multiple times, for which Figure 2.1 was one of those trials. Most plots were broadly similar, and we would expect some variation for each experiment due to the individual Gaussian matrices X that were used for each trial. We empirically found that for approximately every one in ten trials, the left-hand side of Equation (2.24) would exceed the bound imposed by the right. To investigate the tightness of our bound, we have also plotted the ratio of the right-hand side of Equation (2.24) against the

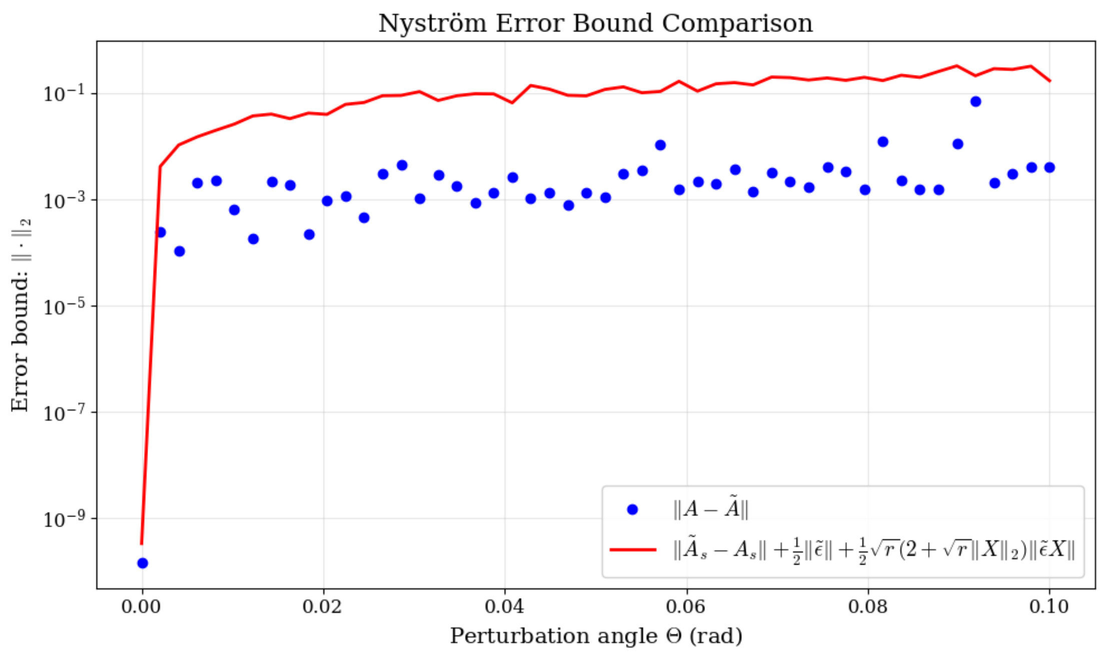


Figure 2.1: A plot showing the numerical implementation of Theorem 2.3. The solid red line shows the upper bound as described in theory, and the blue scattered points are the error between A, an $\tilde{\epsilon}$-near symmetric matrix and its Nyström approximant $\tilde{A}$ in the spectral norm. In all legends, A_{sym} is denoted A_s for legibility.

left-hand side as perturbation angle Θ increases. The results are shown in Figure 2.3. We see that the ratio between the expressions varies greatly from $\mathcal{O}(1)$ to $\mathcal{O}(10^2)$. Note that we have ran this experiment multiple times, and found that the bound ratio rarely exceeds 250. We hypothesise that the reason for such variance in the bound tightness is as a result of the multiple uses of the Cauchy-Schwarz and triangle inequalities we had used in the prior proof.

We were unable to find any simplifications that did not use equations (2.5) and (2.6), so these shall remain. Continuing to assume the validity of the first order approximation of the Neumann series, it follows that the other uses of these inequalities are found in the step from Equation (2.13) to Equation (2.14), followed by another triangle inequality used for $\|I + XC^{\mathrm{T}}\|_\xi \leq \|I\|_\xi + \|XC^T\|_\xi$. To this effect, we shall seek to improve the tightness of these bounds, since the use of the Cauchy-Schwarz inequality is useful in a worst case, where the constituents of the expression D, C and X are not closely linked nor span similar subspaces.

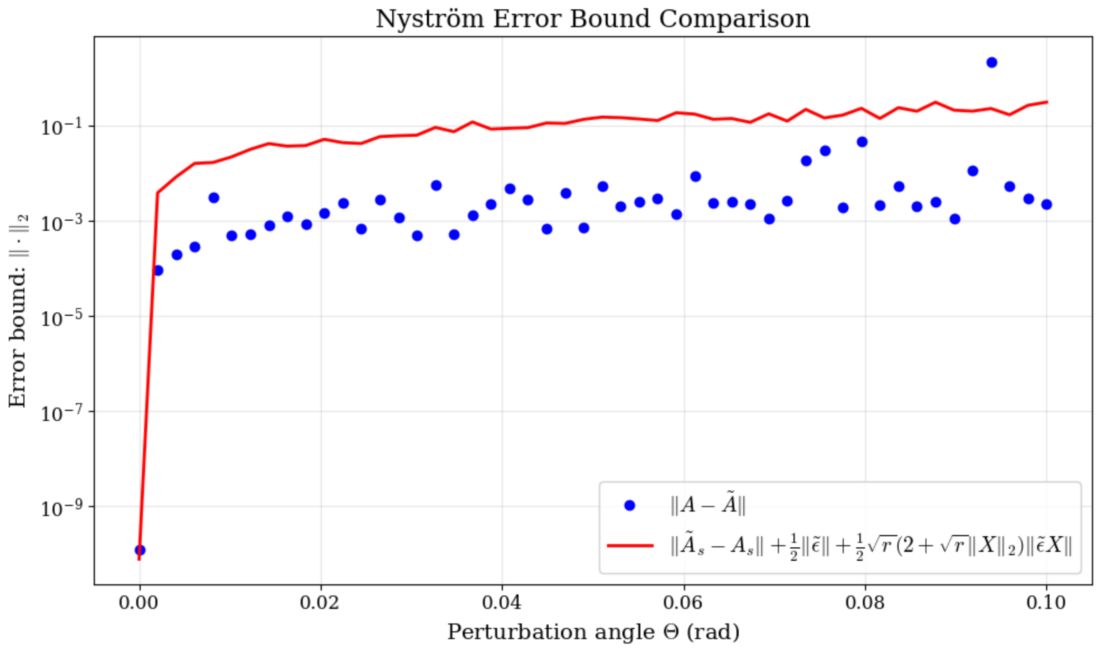


Figure 2.2: A case where Equation (2.24) fails. We see a clear deviated point where $\|A - \tilde{A}\|_2$ protrudes from the line which should bound it from above in accordance with Theorem 2.3.

Remark 2.4. Let us begin by considering Equation (2.13) once again. To proceed, we shall continue to make use of the triangle inequality, though we shall not assume a worst case such as that implied by the Cauchy-Schwarz inequality. We therefore have that

$$\|\tilde{A}_{\text{sym}} - \tilde{A}\|_\xi \lesssim 2\|D\|_\xi + \|DXC^{\mathrm{T}}\|_\xi \tag{2.27}$$

and let us focus now upon $\|D\|_\xi$. We shall begin by considering the expression given in Equation (2.21). Since X is a Gaussian and well-conditioned sketching matrix which is full rank with high probability, and Q_s is an $n \times r$ matrix with orthonormal columns, it follows that since Q_s performs a rotation in the subspace, it does not fundamentally change the overall distribution of the entries of X. What follows is a well-known result that $X^{\mathrm{T}}Q_s$ is also a Gaussian matrix. We seek to find a bound for $\|\tilde{A}_{\text{sym}} - \tilde{A}\|_\xi$, which ideally only contains known artifacts, such as X or the dimension of the problem, and one unknown term, $\tilde{\epsilon}$, the asymmetry of A. As such the difficulty here is that Q_s comes from the QR-decomposition of $A_{\text{sym}}X$, and since we do not have access to A_{sym}, we do not have access to Q_s. From our own empirical findings in our numerical experiments however, we have considered taking another square Gaussian matrix, G, which we shall

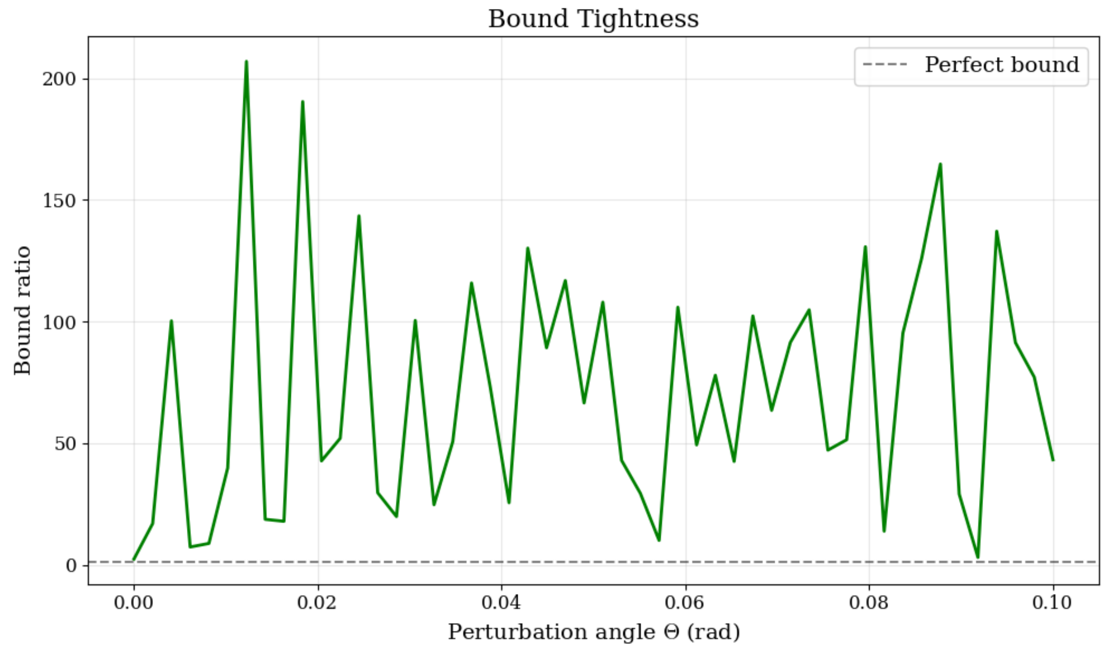


Figure 2.3: A plot showing the quotient of the right-hand side of Equation (2.24) divided by the left-hand side against the perturbation angle Θ, corresponding to the experiment shown in Figure 2.1. For a good bound we would ideally require as small of a coefficient greater than one between the expressions as possible.

assume is of full rank and suitably well-conditioned to computer accuracy, such that we have access to G^{-1}.

We shall then replace $(X^T Q_s)^{-1}$ with G^{-1} with the hypothesis that both span the subspace $\mathbb{R}^r$, and the entries of both $X^{\mathrm{T}} Q_s$ and G are sampled from $\mathcal{N}(0,1)$. We should note this does not have strong theoretical motivations, since the inverse of a Gaussian matrix does not have a well defined distribution of its entries, though it is known that such matrices will have heavier tails than the Gaussian matrices from which they are derived. Upon doing this, we have that in the spectral norm,

$$\|\tilde{A}_{\mathrm{sym}} - \tilde{A}\|_2 \lesssim \|G^{-1}(\tilde{\epsilon} X)^{\mathrm{T}}\|_2 + \frac{1}{2}\|G^{-1}(\tilde{\epsilon} X)^{\mathrm{T}} X C^{\mathrm{T}}\|_2, \tag{2.28}$$

and let us now consider the final term. Rewriting C^{T} in terms of A_{sym} and X, we have that

$$\frac{1}{2}\|G^{-1}(\tilde{\epsilon} X)^{\mathrm{T}} X C^{\mathrm{T}}\|_2 \coloneqq \frac{1}{2}\|G^{-1}(\tilde{\epsilon} X)^{\mathrm{T}} X (X^{\mathrm{T}} A_{\mathrm{sym}} X)^{-1} (A_{\mathrm{sym}} X)^{\mathrm{T}}\|_2. \tag{2.29}$$

Let us now reconsider the QR-decomposition of $A_{\text{sym}}X$, and hence it follows that

$$\frac{1}{2}\|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}X(X^{\mathrm{T}}A_{\text{sym}}X)^{-1}R_s^{\mathrm{T}}Q_s^{\mathrm{T}}\|_2 = \frac{1}{2}\|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}X(X^{\mathrm{T}}A_{\text{sym}}X)^{-1}R_s^{\mathrm{T}}\|_2. \quad (2.30)$$

We seek to remove R_s^{T}, and so let us consider another matrix, $Q_s^T X \in \mathbb{R}^{n\times n}$. Let us assume that $Q_s^{\mathrm{T}}X$ is full rank and hence invertible, since it too as a product of the full column rank matrix Q_s and the Gaussian matrix X. Then we have that $(Q_s^{\mathrm{T}}X)(Q_s^{\mathrm{T}}X)^{-1} = I_r$. Proceeding,

$$\begin{aligned}
&\frac{1}{2}\|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}X(X^{\mathrm{T}}A_{\text{sym}}X)^{-1}R_s^{\mathrm{T}}(Q_s^{\mathrm{T}}X)(Q_s^{\mathrm{T}}X)^{-1}\|_2 \\
&\quad= \frac{1}{2}\|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}X(X^{\mathrm{T}}A_{\text{sym}}X)^{-1}(X^{\mathrm{T}}A_{\text{sym}}X)(Q_s^{\mathrm{T}}X)^{-1}\|_2 &(2.31)\\
&\quad= \frac{1}{2}\|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}X(Q_s^{\mathrm{T}}X)^{-1}\|_2. &(2.32)
\end{aligned}$$

It remains that $Q_s^{\mathrm{T}}X$ is also an unknown matrix. Let us therefore choose another matrix, $\hat{G} \in \mathbb{R}^{r\times r}$, which is assumed to be Gaussian, and suitably invertible. Our hypothesis is now that

$$\|DXC^{\mathrm{T}}\|_2 \simeq \frac{1}{2}\|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}X\hat{G}^{-1}\|_2. \quad (2.33)$$

Putting everything together, it is clear there is no longer a basis to assume the above procedure yields a suitable upper bound, with high probability, and therefore we shall now claim it to be a suitable approximation. In other words, we may write

$$\|\tilde{A}_{\text{sym}} - \tilde{A}\|_2 \simeq \|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}\|_2 + \frac{1}{2}\|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}X\hat{G}^{-1}\|_2 \quad (2.34)$$

for some suitable G and $\hat{G}$ that are accessible to us. As a result, we find the overall expression to be

$$\|A - \tilde{A}\|_2 \simeq \|A_{\text{sym}} - \tilde{A}_{\text{sym}}\|_2 + \|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}\|_2 + \frac{1}{2}\|G^{-1}(\tilde{\epsilon}X)^{\mathrm{T}}X\hat{G}^{-1}\|_2 + \frac{1}{2}\|\tilde{\epsilon}\|_2. \quad (2.35)$$

For comparison, we used the same technique as previously outlined for the numerical results of Theorem 2.3, the results are plotted in Figure 2.4. Over many trials, we determined that the right-hand side of Equation (2.35) often provides a closer alignment with that of the left-hand side compared to Equation (2.24). This leads us to believe that the overzealous use of the Cauchy-Schwarz inequality is a contributing factor of the size of the bound tightness in Figure 2.3. However, as we had previously expressed, we are much less confident in the ability of the right-hand side of Equation (2.35) to act as a good upper bound to the magnitude of $\|A - \tilde{A}\|_2$. Indeed, even in Figure 2.4, there are three violations of the bound, given by the blue scatter points, imposed by the right-hand side of Equation (2.35) by the left-hand side, shown with a solid purple line. We have also plotted the absolute bound tightness in this case, shown in

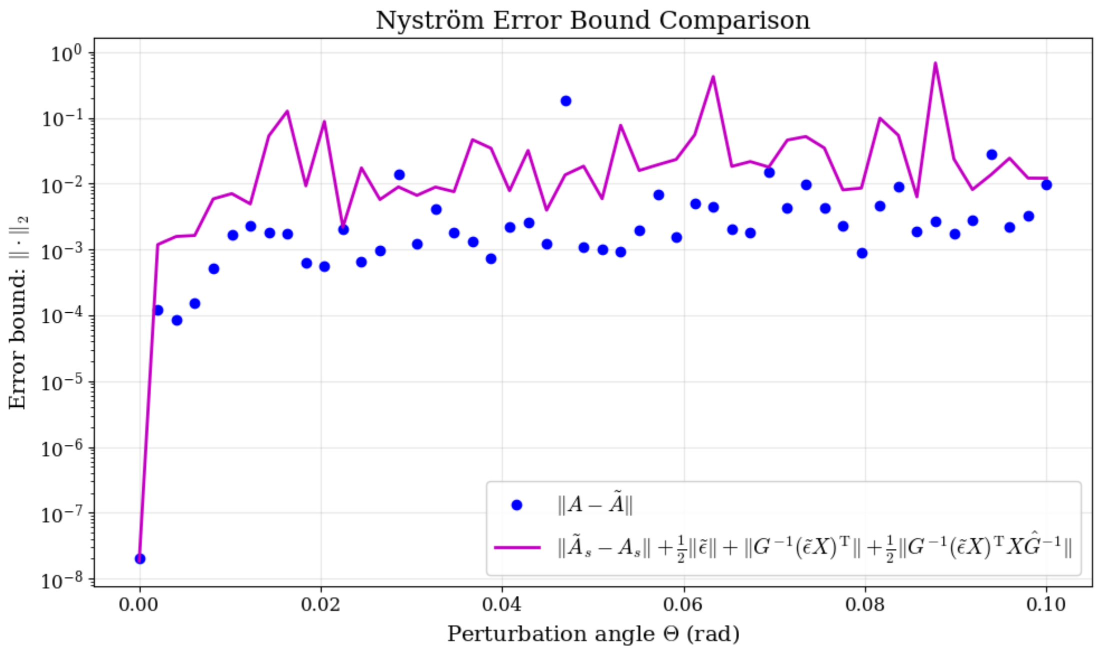


Figure 2.4: A plot of the right-hand side of Equation (2.35) against $\|A - \tilde{A}\|$ in the spectral norm, the former given by a solid purple line and the latter in blue scattered points. We find that often this approximation of the error provides a good estimate of the magnitude of the left-hand side, though with much less certainty about this expression bounding from above.

Figure 2.5. We see that for many points, this method allows the quotient coefficient to remain below 50, significantly better than what we see in Figure 2.3. However, this method is also prone to larger spikes. We had previously mentioned that from our experiments, it was extremely rare that such disparities in the size of the bound and numerical results of Theorem 2.3 exceed 250. In the case of Equation (2.35), we find that the spikes exhibited can be much larger in magnitude. Though uncommon, successive trials did eventually yield spikes of $\mathcal{O}(10^4)$ in magnitude, significantly larger than those found in Figure 2.3, which consistently stay in $\mathcal{O}(10)$–$\mathcal{O}(10^2)$.

For consistency in evaluating the relative strengths of both procedures, we had decided to run the experiment again ten times for both expressions. For each trial,

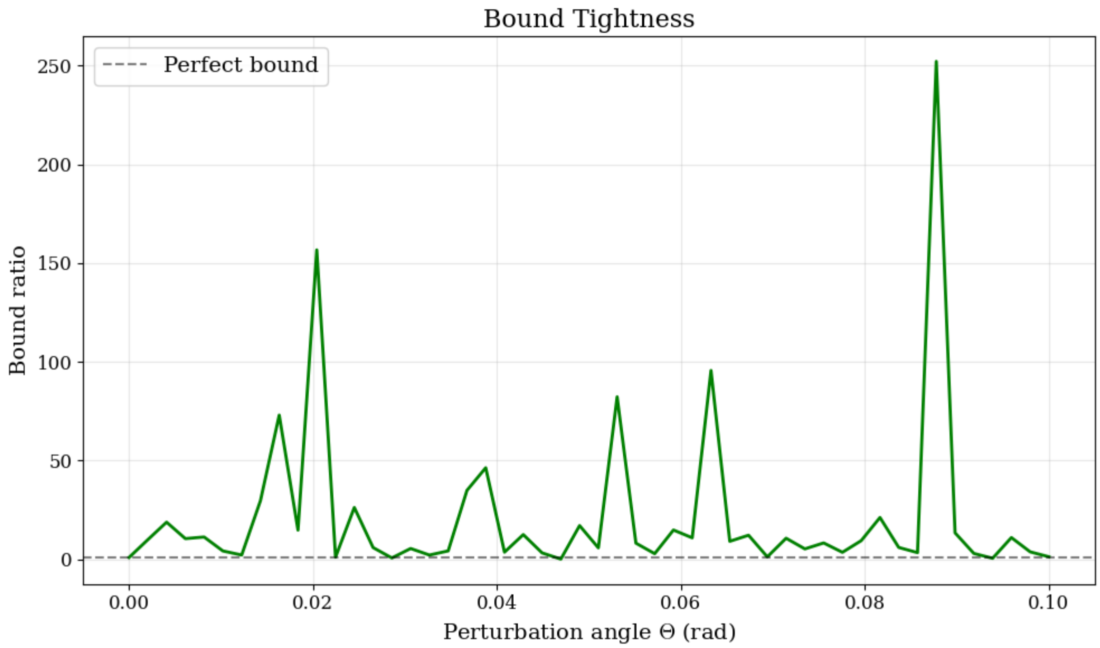


Figure 2.5: A plot of the absolute bound tightness corresponding to Figure 2.4. We see that for many cases, the coefficient between the left- and right-hand sides is smaller compare to the complementary plot in Figure 2.3, although this procedure is much more prone to larger spikes.

we would add the minimum, the maximum, the mean and the median values for the ratio bound calculated. In other words, where each method yielded the best bound or estimate, its worst bound or estimate, and the average bound or estimate. We had decided upon the use of both the mean and median to this end, as each average would implicitly favour one method over the other. Since the right-hand side of Equation (2.24) is more consistently an upper bound for the left-hand side, and from our previous experiments, we see this expression is more consistent with spikes of smaller magnitude, then the mean expression would be more advantageous. For the estimation given in Equation (2.35), it should be clear from Figure 2.5 that we expect a tighter bound in many iterations, but with considerably larger spikes in others. Spikes of very large magnitudes would of course increase the mean value more disproportionately than others, and so we use the median in order to remove those extreme values when they occur.

For Table 2.1, we see that minimum value was consistently of $\mathcal{O}(1)$. Since each trial contained a range of fifty values of perturbation angle Θ, we had noted that there is a small but non-negligible probability that the left-hand side exceeds the right-hand side for some cases, yielding a ratio less than one. Even when this happens, we see that the resulting value is of size 0.8–0.99. In other words, when Theorem 2.3 fails, it usually is only in the vicinity of 1–25%. As we had previously mentioned, the maximum value is the most pessimistic estimation of $\|A - \tilde{A}\|_2$. We see again that the maxima are consistent across the ten trials and the fifty angles in each trial. We also see that they remain mostly below a factor of 250. The mean and median results are always of comparable size and reflect the uniformity of spikes occurring across all values. As a result, both the underestimations and overestimations counteract each other and we find a very stable coefficient quotient magnitude consistently in the range 50–65.

Trial	Minimum	Maximum	Mean	Median
1	0.99	165.27	57.89	48.33
2	0.81	232.14	69.70	50.10
3	0.80	186.73	70.99	60.09
4	1.39	180.77	64.30	49.34
5	0.83	249.59	67.64	61.10
6	0.89	222.35	67.52	52.61
7	0.61	216.32	70.32	65.47
8	0.83	207.14	70.12	55.68
9	0.93	315.68	75.09	64.05
10	0.35	223.64	66.39	57.86

Table 2.1: A table capturing the coefficient sizes of the quotient between the right-hand side and left-hand side of Equation (2.24), to two decimal places. We find for Theorem 2.3, these results are of consistent magnitude for each column.

Table 2.2 shows the results of the next test, this time using the quotient of the left- and right-hand sides of Equation (2.35) in order to calculate the same values. As we would expect, the first column of minima is often smaller in magnitude than those found in Table 2.1. This is because this expression was more focused on being an estimation of the magnitude of $\|A - \tilde{A}\|_2$ rather than a more strict upper bound, as was our aim for Theorem 2.3. In this case, the minimum was always within $\mathcal{O}(10)$ of the true value in the most pessimistic cases, and within $\mathcal{O}(1)$ in the better cases.

The column of maxima shows us that the magnitude of the largest spikes is much harder to estimate compared the previous method, we do in fact see the overestimation ranging from $\mathcal{O}(10^2)$ to $\mathcal{O}(10^4)$ as we had previously stated. However, we should note that although we are comparing the magnitude of these overestimations compared to those found in Table 2.1, we have not yet considered the frequency of occurrence of these spikes. Indeed, Figure 2.3 shows us a typical graph output by the first procedure. We see many spikes of similar size with consistent frequency throughout the range of perturbation angles. This is not the case for Figure 2.5. Though the larger spikes here are of comparable magnitude, they occur with much lower frequency to those in Figure 2.3. Speaking empirically, even for spikes of size $\mathcal{O}(10^4)$, we only ever see a very small number of outlying spikes of overestimation. We see a more significant

Trial	**Minimum**	**Maximum**	**Mean**	**Median**
1	0.39	398.22	29.62	5.82
2	0.31	836.09	35.43	4.75
3	0.03	1033.12	44.43	7.35
4	0.06	668.20	35.64	8.74
5	0.37	11795.23	310.76	7.65
6	0.51	2061.32	115.96	8.50
7	0.18	517.38	38.26	6.41
8	0.33	2413.03	133.01	10.29
9	0.07	1041.05	59.21	5.58
10	0.17	324.36	22.57	6.22

Table 2.2: A table capturing the coefficient sizes of the quotient between the right-hand side and left-hand side of Equation (2.35), to two decimal places. We find for this case, there is much greater variance in the magnitudes for each column, though both averages yield some promising results.

divergence in values between the mean and median averages across Table 2.2. As we would expect, the mean can be greatly influenced by spikes of extremely large magnitude. As we had previously mentioned however, the fact that such spikes appear much less often than in the previous case, it follows that the use of the median average would be much more beneficial for error estimation. We should note however, that even for most entries in the mean column, where the maximum stays in the vicinity of $\mathcal{O}(10^2)$, the mean of the experiments here are lower than or equal to the mean of experiments recorded in Table 2.1. Nevertheless, it is clear from all of our experiments that the median column of Table 2.2 is significantly less than both the median column of Table 2.1 and its own mean column, since with this method we can easily remove the poor approximations on either side of the $\|A-\tilde{A}\|_2$ points.

2.3 Effects of Oversampling $\tilde{\epsilon}$-Near-Symmetric Matrices

Returning to Theorem 2.3, one should ask whether any of our prior analysis above can be used in the case where X is an oversampling matrix, that is where $X \in \mathbb{R}^{n\times s}$, for $s > \operatorname{rank}(A) = r$. It should be clear that our previous work applies to any low rank approximation to matrix A in the case where $X \in \mathbb{R}^{n\times s}$ and $s < \operatorname{rank}(A)$. To answer this, let us now inspect the case where X^TAX is rank deficient, say $X \in \mathbb{R}^{n\times s}$ where $s > r$, we seek to find an analogue for the Neumann series in the case of a pseudoinverse. For brevity, denote $M \coloneqq X^{\mathrm{T}}A_{\mathrm{sym}}X$ and $E \coloneqq \frac{1}{2}X^{\mathrm{T}}\tilde{\epsilon}X$. Now, let us denote $\Pi = MM^{\dagger}$. Then, we find in Stewart [16] that for $\|M^{\dagger}\|_2^2\|\Pi E\Pi\|_2 < 1$, it follows that $\operatorname{rank}(M+E) = \operatorname{rank}(M)$, and hence to first order, we have that

$$(M+E)^{\dagger} = M^{\dagger} - M^{\dagger}\,\Pi E\Pi\, M^{\dagger} - (M^{\dagger})^2\,\Pi E\Pi + \Pi E\Pi\,(M^{\dagger})^2 \;+\; \mathcal{O}(\|E\|^2). \quad (2.36)$$

Set

$$\begin{aligned} F \coloneqq \Big(Y+Z\Big)\,(M+E)^{\dagger}\,\Big(Y+Z\Big)^{\mathrm{T}} &= \\ &\Big(\Big(A_{\mathrm{sym}}+\tfrac{1}{2}\tilde{\epsilon}\Big)\,X\Big)\,\Big(X^{\mathrm{T}}A_{\mathrm{sym}}X+\tfrac{1}{2}X^{\mathrm{T}}\tilde{\epsilon}X\Big)^{\dagger}\,\Big(\Big(A_{\mathrm{sym}}+\tfrac{1}{2}\tilde{\epsilon}\Big)\,X\Big)^{\mathrm{T}}. \end{aligned} \quad (2.37)$$

Using Equation (2.36),

$$\begin{aligned} F = {} & (Y+Z)\,M^{\dagger}\,(Y+Z)^{\mathrm{T}} && \text{(i)} \\ & - (Y+Z)\,M^{\dagger}\,\Pi E\Pi\,M^{\dagger}\,(Y+Z)^{\mathrm{T}} && \text{(ii)} \\ & - (Y+Z)\,(M^{\dagger})^2\,\Pi E\Pi\,(Y+Z)^{\mathrm{T}} && \text{(iii)} \\ & + (Y+Z)\,\Pi E\Pi\,(M^{\dagger})^2\,(Y+Z)^{\mathrm{T}} \;+\; \mathcal{O}(\|E\|_2^2). && \text{(iv)} \end{aligned} \quad (2.38)$$

Expanding each line of Equation (2.38), where the Roman numerals correspond to each line of the expression:

$$
\begin{aligned}
&\text{(i)} \quad (Y+Z)\,M^{\dagger}\,(Y+Z)^{\mathrm{T}} \\
&\qquad = \underbrace{YM^{\dagger}Y^{\mathrm{T}}}_{\tilde{A}_{\text{sym}}} + YM^{\dagger}Z^{\mathrm{T}} + ZM^{\dagger}Y^{\mathrm{T}} + ZM^{\dagger}Z^{\mathrm{T}}, \\
&\text{(ii)} \quad -(Y+Z)\,M^{\dagger}\,\Pi E \Pi\,M^{\dagger}\,(Y+Z)^{\mathrm{T}} \\
&\qquad = -YM^{\dagger}\Pi E\Pi M^{\dagger}Y^{\mathrm{T}} - YM^{\dagger}\Pi E\Pi M^{\dagger}Z^{\mathrm{T}} \\
&\qquad\quad - ZM^{\dagger}\Pi E\Pi M^{\dagger}Y^{\mathrm{T}} - ZM^{\dagger}\Pi E\Pi M^{\dagger}Z^{\mathrm{T}}, \\
&\text{(iii)} \quad -(Y+Z)\,(M^{\dagger})^2\,\Pi E\Pi\,(Y+Z)^{\mathrm{T}} \\
&\qquad = -Y(M^{\dagger})^2\Pi E\Pi Y^{\mathrm{T}} - Y(M^{\dagger})^2\Pi E\Pi Z^{\mathrm{T}} \\
&\qquad\quad - Z(M^{\dagger})^2\Pi E\Pi Y^{\mathrm{T}} - Z(M^{\dagger})^2\Pi E\Pi Z^{\mathrm{T}}, \\
&\text{(iv)} \quad (Y+Z)\,\Pi E\Pi\,(M^{\dagger})^2\,(Y+Z)^{\mathrm{T}} \\
&\qquad = \; Y\Pi E\Pi(M^{\dagger})^2Y^{\mathrm{T}} + Y\Pi E\Pi(M^{\dagger})^2Z^{\mathrm{T}} \\
&\qquad\quad + Z\Pi E\Pi(M^{\dagger})^2Y^{\mathrm{T}} + Z\Pi E\Pi(M^{\dagger})^2Z^{\mathrm{T}}.
\end{aligned}
$$

Since $Z = \mathcal{O}(\|\tilde{\epsilon}\|_2)$ and $E = \mathcal{O}(\|\tilde{\epsilon}\|_2)$, F is, to first order:

$$\tilde{A}_{\text{sym}} + YM^{\dagger}Z^{\mathrm{T}} + ZM^{\dagger}Y^{\mathrm{T}} - YM^{\dagger}\Pi E\Pi M^{\dagger}Y^{\mathrm{T}} + Y\Big[\Pi E\Pi,\,(M^{\dagger})^2\Big]Y^{\mathrm{T}}$$

and every remaining term shown above is $\mathcal{O}(\|\tilde{\epsilon}\|_2^2)$. Here $[\Pi E\Pi, (M^{\dagger})^2]$ denotes the commutator $\Pi E\Pi(M^{\dagger})^2 - (M^{\dagger})^2\Pi E\Pi$. It follows that

$$\|\tilde{A}-\tilde{A}_{\text{sym}}\|_2 \simeq \|YM^{\dagger}Z^{\mathrm{T}}+ZM^{\dagger}Y^{\mathrm{T}}-YM^{\dagger}\Pi E\Pi M^{\dagger}Y^{\mathrm{T}}+Y\Big[\Pi E\Pi,\,(M^{\dagger})^2\Big]Y^{\mathrm{T}}\|_2. \tag{2.39}$$

The first term, $YM^{\dagger}Z^{\mathrm{T}}$, cannot be simplified since

$$YM^{\dagger}Z^{\mathrm{T}} := (A_{\text{sym}}X)(X^{\mathrm{T}}A_{\text{sym}}X)^{\dagger}\left(\frac{1}{2}\tilde{\epsilon}X\right)^{\mathrm{T}}, \tag{2.40}$$

and so by following the same steps as for the invertible case, we find that

$$\|YM^{\dagger}Z^{\mathrm{T}}\|_2 = \|(X^{\mathrm{T}}Q_s)^{\dagger}(X^{\mathrm{T}}A_{\text{sym}}X)(X^{\mathrm{T}}A_{\text{sym}}X)^{\dagger}\left(\frac{1}{2}\tilde{\epsilon}X\right)^{\mathrm{T}}\|_2. \tag{2.41}$$

Since we cannot impose that $\Pi := MM^{\dagger} = I_s$, this cannot be simplified further. For the remaining four terms, we have a similar problem, namely that there is no simplification of the $M^\dagger Y^{\mathrm{T}}$ term in general. Without other restricting assumptions, we cannot bound or approximate the magnitude of $\|\tilde{A} - \tilde{A}_{\mathrm{sym}}\|_2$ without the imposition of stricter assumptions. Indeed, one would need access to $Y, Y^{\mathrm{T}}, Z, Z^{\mathrm{T}}, M, M^\dagger, \Pi$ and E in this case, as opposed to only $\|X\|_\xi$ and $\|\tilde{\epsilon}X\|_\xi$ in the invertible case.

2.4 Analysis of $\tilde{\epsilon}$-Near-Symmetry

Considering the broader scope of our research, it is helpful to ask whether the work we have done is of any practical use. Indeed, one may regard Theorem 2.3 to be an algebraic 'sleight of hand', whereby we have simply transferred the complexity of $\|A - \tilde{A}\|_\xi$ into two parts: the accuracy of the Nyström approximation, and the asymmetry of the matrix, $\tilde{\epsilon}$. All of our work thus far has been predicated on the bound of $\tilde{\epsilon}$ to be known. This is troubling since we had previously stated that we have access to neither A nor A^{T}, and so to the most pessimistic, our work may even be regarded as futile.

We shall try to address all of these questions now, by examining the properties of $\tilde{\epsilon}$, drawing similarities with the notion of ϵ-near-symmetric matrices as described in Boullé, Halikias, *et al.* [2], as well as their differences. It is our hope that can bound the size of $\tilde{\epsilon}$ for some near-symmetric matrices.

Let us as before, consider a matrix $A \in \mathbb{R}^{n\times n}$ to be $\tilde{\epsilon}$-near-symmetric and we shall assume $\mathrm{rank}(A) = r$. Then, we have that $\tilde{\epsilon} = A - A^{\mathrm{T}}$. Let us assume the full SVD of $A = U_{A,f}\Sigma_{A,f}V_A^{\mathrm{T}}$. Then, we shall have

$$A = \underbrace{\left[\begin{array}{c:c} U_A & U^\perp \end{array}\right]}_{U_{A,f}} \underbrace{\left[\begin{array}{c:c} \Sigma_r & 0_{r\times(n-r)} \\ \hdashline 0_{(n-r)\times r} & 0_{(n-r)\times(n-r)} \end{array}\right]}_{\Sigma_{A,f}} \underbrace{\left[\begin{array}{c} V_A^{\mathrm{T}} \\ \hdashline V_\perp^{\mathrm{T}} \end{array}\right]}_{V_{A,f}^{\mathrm{T}}}. \tag{2.42}$$

We shall define $V_{A,f}$ to be a perturbation of $U_{A,f}$. We have denoted $U_{A,f} = [u_1 \cdots u_n]$ to be the orthonormal columns of $U_{A,f}$, and $V_{A,f} = [v_1 \cdots v_n]$ to be the orthonormal columns of $V_{A,f}$. As before, for each column v_i, for $i \in 1, \ldots, r$, we shall say

$$v_i = u_i \cos(\theta_i) + w \sin(\theta_i), \text{ for } w \perp \mathrm{span}(U_A),$$

for $\|w\|_2 = 1$. In other words, we can have that $w \in \mathrm{span}(U^\perp)$. Returning to the full matrix expression, we shall write

$$\tilde{\epsilon} := A - A^{\mathrm{T}} = U_{A,f}\Sigma_{A,f}V_{A,f}^{\mathrm{T}} - V_{A,f}\Sigma_{A,f}U_{A,f}^{\mathrm{T}}, \tag{2.43}$$

and we shall consider the magnitude of $\tilde{\epsilon}$ in the spectral and Frobenius norms. We have that

$$\begin{aligned}
\|\tilde{\epsilon}\|_\xi &= \|U_{A,f}\Sigma_{A,f}V_{A,f}^{\mathrm{T}} - V_{A,f}\Sigma_{A,f}U_{A,f}^{\mathrm{T}}\|_\xi && (2.44)\\
&= \|\Sigma_{A,f}V_{A,f}^{\mathrm{T}} - U_{A,f}^{\mathrm{T}}V_{A,f}\Sigma_{A,f}U_{A,f}^{\mathrm{T}}\|_\xi && (2.45)\\
&= \|\Sigma_{A,f} - U_{A,f}^{\mathrm{T}}V_{A,f}\Sigma_{A,f}U_{A,f}^{\mathrm{T}}V_{A,f}\|_\xi. && (2.46)
\end{aligned}$$

Let us now consider matrix $U_{A,f}^{\mathrm{T}}V_{A,f}$, and for simplicity, let us assume that only one arbitrary column, say the k^{th} column of $U_{A,f}$ has been perturbed, for some $1 \leq k \leq r$. Then, without loss of generality, we have that

$$V_{A,f} = [u_1 \cdots u_k \cos(\theta_k) + w\sin(\theta_k) \cdots u_r \cdots u_n] \tag{2.47}$$

when written column-wise. By simple computation, one should easily see that we have

$$U_{A,f}^T V_{A,f} = \begin{bmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cos(\theta_k) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 1 \end{bmatrix}. \tag{2.48}$$

From here, it follows that

$$U_{A,f}^T V_{A,f}\Sigma_{A,f}U_{A,f}^T V_{A,f} = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_k\cos^2(\theta_k) & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & \sigma_r & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix} \tag{2.49}$$

and hence

$$\Sigma_{A,f} - U_{A,f}^{\mathrm{T}} V_{A,f} \Sigma_{A,f} U_{A,f}^{\mathrm{T}} V_{A,f} = \begin{bmatrix} 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_k \sin^2(\theta_k) & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix}. \tag{2.50}$$

It follows that in this case, in the spectral norm, we have $\|\tilde{\epsilon}\|_2 = \sigma_k \sin^2(\theta_k)$. What is particularly pleasing here is that for $\tilde{\epsilon}$-near-symmetric matrices, we find that $\|\tilde{\epsilon}\|_2 \simeq \sigma_k \theta_k^2$, that is, we have the same perturbation angle dependency up to constant coefficients as the notions of ϵ-near-symmetry as considered in Boullé, Halikias, *et al.* [2]. For more general $\tilde{\epsilon}$-near-symmetric matrices, where each column of $V_{A,f}$ is perturbed by some individual perturbation angle θ_k for each k, we trivially see in the spectral norm:

$$\|\tilde{\epsilon}\|_2 = \max_{1 \leq k \leq r} \sigma_k \sin^2(\theta_k), \tag{2.51}$$

and again for $\tilde{\epsilon}$-near-symmetric matrices, we have $\|\tilde{\epsilon}\|_2 = \max_{1 \leq k \leq r} \sigma_k \theta_k^2$. What is assuring here is the rediscovery of the imposition of the 'column-wise near-symmetry' we had spoken of in Equation (1.30), again relating to the near-symmetry of the matrix through the greatest perturbation across all columns.

We demonstrate our findings through numerical experiments corresponding to Figure 2.6. We begin by forming an $\tilde{\epsilon}$-near-symmetric matrix in the way previously outlined; by taking an eigendecomposition of a positive definite symmetric matrix, and perturbing the right eigenspace, hence forming the SVD. In this case, we had made $V_{A,f}$ by perturbing columns of $U_{A,f}$. We had ensured that this was done for a matrix with poor conditioning, to prove the concept although we accept that very poorly conditioned matrices are not necessarily desirable for use in the Nyström approximation. Each column was individually perturbed for $\Theta = 0$–$\frac{\pi}{2}$ rad. We chose this range because we would be able to hypothesise that on a logarithmic plot, our plotted points would be joined by a line of gradient 2 for small angles, before levelling out at larger angles and reaching a maximum for $\sin^2(\Theta) = 1$. We had taken ten singular values between ranges of 10^{-6} and 10^2 which were of exponentially decreasing magnitude along the diagonal of $\Sigma_{A,f}$, with the dark blue line in Figure 2.6 representing the perturbation of the right singular vector corresponding to the largest of those singular values, and the lowest pink line corresponding to the perturbed right singular vector with the smallest non-zero singular value.

We had previously outputted the average gradient for each line for small perturbation angles, which we have since removed for improved legibility. We did this using the `scipy.stats.linregress()` program available in the SciPy Python library (see Virtanen, Gommers, *et al.* [19]). All lines output a gradient between 1.98–1.99. Figure 2.6 also includes horizontal dotted lines, each dotted line represents the singular value corresponding to the perturbed singular vector, and so we have given them the same colour as our plotted lines. We see at larger values, all lines begin to flatten, reaching a maximum at $\Theta = \frac{\pi}{2}$. From this, we conclude that for $\tilde{\epsilon}$-near-symmetric matrices formed through the method above, we can bound the size of $\|\tilde{\epsilon}\|_2$ by the singular value corresponding to the perturbed singular vector.

Lastly, the black line above represents the perturbation of all right singular vectors by the same angle. One would usually expect this line to overlap entirely with the blue line, as this would be what our analysis would suggest, however for some $\tilde{\epsilon}$-near-symmetric matrices, we show that this is not guaranteed. The problem is as a result of how we formed the perturbation of the columns of $V_{A,f}$, in particular, how we had defined w. In our experiment, we had taken any $w \perp \operatorname{span}(U_A)$. Therefore, considering the thin SVD of A, we can write

$$V_A = U_A \cos(\Theta) + w\mathbf{1}_r^{\mathrm{T}} \sin(\Theta), \tag{2.52}$$

where we shall denote $\mathbf{1}_r$ to be the vector $[1 \cdots 1]^{\mathrm{T}} \in \mathbb{R}^r$. We still have that $U^{\mathrm{T}}U = I_r$, although now, we have that

$$V_A^{\mathrm{T}} V_A = I_r \cos^2(\Theta) + \mathbf{1}_r \mathbf{1}_r^{\mathrm{T}} \sin^2(\Theta). \tag{2.53}$$

Our code had used `numpy.linalg.qr()` in order to maintain stable numerical orthogonality, even though this should have been preserved through our derivation theoretically. The problem arises from using the same w to perturb each column of U_A, thereby breaking the orthogonality between the columns. Let $V_A = V_0 R_0$ be the QR-decomposition of V_A. It follows that R_0 is invertible, since V_A is of full column

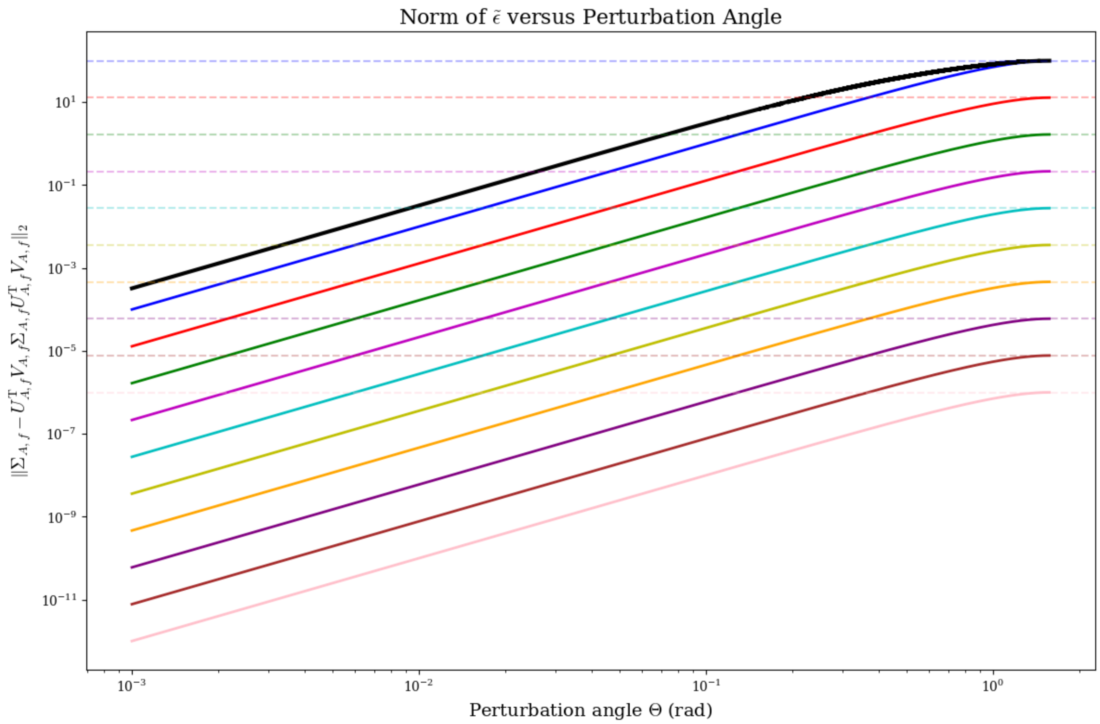


Figure 2.6: A plot to show how $\|\tilde{\epsilon}\|_2$ increases with perturbation angle θ_i for every column i. Here, we have taken $n = 20$ and $r = 10$. Each of the coloured lines correspond to a singular value, the distribution of which is given by `numpy.logspace(-6,2,r)` to ensure good legibility. The black line represents how $\|\tilde{\epsilon}\|_2$ changes for all columns perturbed by the same overall perturbation angle, Θ.

rank, and substituting, we may write $U_A^{\mathrm{T}} V_0 = U_A^{\mathrm{T}} V_A R^{-1}$. We also have that

$$V_A^{\mathrm{T}} V_A = \begin{bmatrix} 1 & \sin^2(\Theta) & \cdots & \sin^2(\Theta) \\ \sin^2(\Theta) & 1 & \cdots & \sin^2(\Theta) \\ \vdots & \vdots & \ddots & \vdots \\ \sin^2(\Theta) & \sin^2(\Theta) & \cdots & 1 \end{bmatrix} = R_0^{\mathrm{T}} R_0, \tag{2.54}$$

a Cholesky factorisation. We also have that $U_A^{\mathrm{T}} V_A$ remains equal to $I_r \cos(\Theta)$. It follows that $U^{\mathrm{T}} V_0 = \cos(\Theta) R_0^{-1}$, and the singular values of R_0^{-1} can be given by $\frac{1}{\sqrt{\sigma_i(R_0^{\mathrm{T}} R_0)}}$, for each i. Focusing on Equation (2.53), we see it is a rank-1 update of $I_r \cos^2(\Theta)$. Let us consider some vector x, where $\mathbf{1}_r^{\mathrm{T}} x = 0$. It follows that $\cos^2(\Theta)$ is an eigenvalue of $V_A^{\mathrm{T}} V_A$ with an algebraic multiplicity of $r - 1$. Let us now consider the vector $\mathbf{1}_r$, Then, $(\mathbf{1}_r \mathbf{1}_r^{\mathrm{T}}) \mathbf{1}_r = r \mathbf{1}_r$. It follows that $V_A^{\mathrm{T}} V_A \mathbf{1}_r = (\cos^2(\Theta) + r \sin^2(\Theta)) \mathbf{1}_r$, and so

$\cos^2(\Theta) + r\sin^2(\Theta)$ is the final eigenvalue. As a result, the singular values of $U_A^{\mathrm{T}} V_0$ are given by 1, with multiplicity $r-1$, and $\frac{\cos(\Theta)}{\sqrt{\cos^2(\Theta)+r\sin^2(\Theta)}}$, with multiplicity 1. In actuality therefore, we have rotated the subspace $\mathrm{span}(V_A)$ away from $\mathrm{span}(U_A)$ by only one non-zero canonical angle which we shall denote as ϕ:

$$\phi = \cos^{-1}\left(\frac{\cos(\Theta)}{\sqrt{\cos^2(\Theta) + r\sin^2(\Theta)}}\right). \tag{2.55}$$

By plotting both expressions, we see that

$$\sin^2(\Theta) \leq \sin^2(\phi), \tag{2.56}$$

in the domain $[0, \frac{\pi}{2}]$, with equality exactly only at points 0 and $\frac{\pi}{2}$, as we would expect theoretically and as we would expect from Figure 2.6.

To address this, we shall make an adjustment to the way we form the columns of V_A. Let us again write $U_A = [u_1 \cdots u_r]$ to be the left singular vectors of A, from which we shall form $V_A = [v_1 \cdots v_r]$, the right singular vectors. For each column, let v_i be given by

$$v_i = u_i \cos(\theta_i) + w_i \sin(\theta_i), \text{ for } 1 \leq i \leq r, \ \|w_i\|_2 = 1, \tag{2.57}$$

where now, we shall define every $w_i \perp \mathrm{span}(U_A)$ and also $w_i \perp w_j$, for all $i \neq j$. In other words, we shall give every column of V_A a perturbation that is orthogonal to the column-space of U_A, but also to every other column of V_A to ensure our columns remain orthonormal to each other by construction. To procure such a set of columns w_i and maintain orthogonality, it is logical to use sample each of the columns of $U_\perp$, or $\{u_{r+1}, \ldots, u_n\}$ of $U_{A,f}$. Here we reach a dimensional problem; in accordance with the rank-nullity theorem, we must ensure that $r \leq \lfloor \frac{n}{2} \rfloor$ in order to ensure there are enough columns of $U_\perp$ to guarantee there exist enough such w_i of each column to maintain orthogonality. This is of course, highly likely for most applications of the Nyström method.

2.5 Operator Learning: Nyström versus Neural Networks

In this section, we shall begin to discuss the utility of our work, in particularly by comparing the Nyström approximation with other methods commonly used in operator learning, most notably, neural networks. The most significant advantage that neural networks possess against the Nyström approximation is their ability to learn non-symmetric operators, and so our goal shall be to trail both methods on $\tilde{\epsilon}$-near-symmetric matrices. We shall begin by introducing the foundations of how a neural network works, before explaining how we had used a neural network to learn a near-symmetric matrix. We shall then present some of our experimental results before drawing some conclusions from our work.

2.5.1 Neural Networks: A Brief Introduction

A neural network is a mathematical object that is used to recover functions, or in our case, linear operators. They are formed from interconnected nodes that span from the input to the output, known as neurons, due to the clear likeness between these nodes and neurons in biology. Indeed, much of the vocabulary used here is derived from attempts to model biological neural networks artificially (see McCulloch and Pitts [12]). Neurons in an artificial neural network are arranged in layers, with information passing from the input layer, through intermediate layers, known as hidden layers, to the output layer. The signal input to each neuron is a linear combination of the outputs of the previous layers, in other words, the input into an n^{th} layer neuron is a linear combination of the outputs of the $n-1^{th}$ layer neurons. The output of each neuron is governed by its corresponding activation function, the function that calculates an output of the node depending on its inputs. Non-linear problems can be solved if the activation functions are non-linear. Figure 2.7 shows a diagram of a neural network.

Before we introduce our findings, we should define each of the terms that we shall use. Every neuron in a neural network receives an input and returns an output. The activation function, $\mathscr{F}$, governs what is returned in order to maximise the learning of the neural network from the data. Most neural networks use non-linear activation functions, such as ReLU, given by $\mathscr{F}(\mathsf{z}) \coloneqq \max(0, \mathsf{z})$, for some input . For example, consider Figure 2.7 and consider i^{th} hidden layer, for $i \leq L$. Let $\mathsf{y}_1, \ldots, \mathsf{y}_{l_{k-1}}$ be the inputs into some hidden node, $H^{(i)}_{l_k}$. Each neuron assigns weights their inputs relative to how well they minimise error. Let us denote the weighting functions to be given by $\mathsf{w}_1, \ldots \mathsf{w}_{l_{k-1}}$. Then, putting this together, the output of the neuron is given by

$$H^{(i)}_{l_k}(\mathsf{y}_1, \ldots, \mathsf{y}_{l_{k-1}}) = \mathscr{F}\left(\sum_{j=1}^{l_{k-1}} \mathsf{w}_j(\mathsf{y}_j) + \mathsf{b}_j\right), \tag{2.58}$$

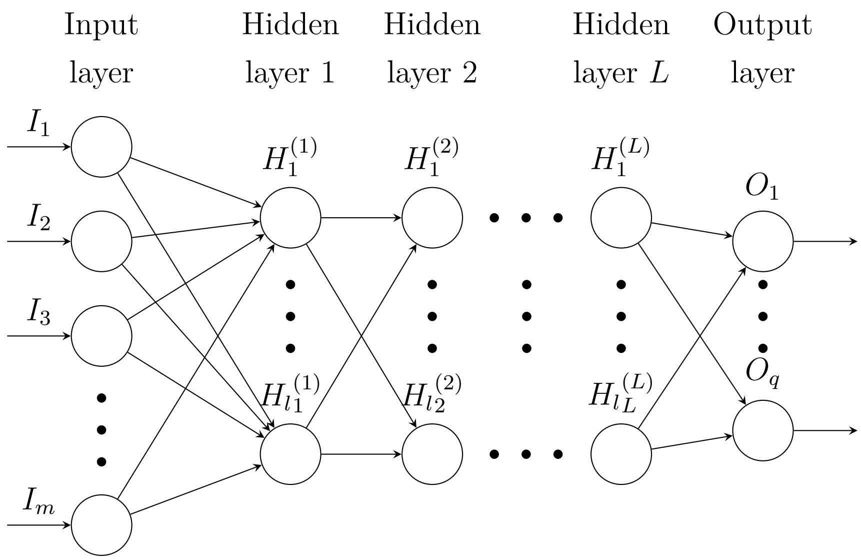


Figure 2.7: A schematic of a general neural network architecture with m neurons in the input layer, followed by L hidden layers, each with l neurons. The final layer consists of q neurons that return the output.

for some added biases b_j. We shall now discuss some more general properties of activation functions, and we shall start with non-linearity. When the activation function is non-linear, then a two-layer neural network can be proven to be a *universal function approximator*. This is a corollary of the universal approximation theorem [8], which states that given a family of neural networks of sufficient depth, for each function g in a given function space, there exists a sequence of neural networks $\{\gamma_1, \ldots, \gamma_n, \ldots\}$ from this family, such that $\gamma_n \to f$. That is, the family of neural networks are *dense* in the function space, meaning that the use of set $\{\gamma_1, \ldots, \gamma_n\}$ can either recreate function g exactly, or approximate it to arbitrary accuracy. This serves as a limit theorem and only proves that there exist sequences $\{\gamma_1, \gamma_2, \ldots\}$ that map to g, although it provides no algorithm for finding such a sequence, only guaranteeing that one exists.

Neural networks learn through the optimisation of each weight w_j and each bias b_j in each node. This optimisation is encoded by the *loss function*. If we take training data along with its known desired output data, we can train a neural network by inputting the training data, taking the output of the network and comparing it to the desired output we were expecting. In the cases of operator learning, we are often interested in regression loss functions, that is those that have continuous outputs. Common loss functions are mean squared error and mean absolute error.

This error is minimised for continuously differentiable activation functions through gradient descent algorithms, such as backpropagation. Backpropagation relies on the continuity and existence of the derivates of the outputs, since backpropagation uses the chain rule of calculus in order to reverse engineer where the errors are coming from. Once the respective biases and weights are readjusted, we can then input the training data into the neural network again, measure the loss function and use gradient descent once again in order to further improve the weighting and bias values for each neuron.

2.5.2 Numerical Experiments with a Neural Network

We begin considering a symmetric positive semi-definite matrix $A_{\text{sym}} \in \mathbb{R}^{n\times n}$, whose thin eigendecomposition is given by $U_A \Sigma_A U_A^{\mathrm{T}}$ as before. For our experiments, we had taken $n = 100$ and $r = 10$, where $\text{rank}(A_{\text{sym}}) = r$. It follows that $U_A \in \mathbb{R}^{n\times r}$. We had generated the singular values through `numpy.logspace(-3, 2, r)[::1]`, that is, ten singular values in exponentially descending order, which form the major diagonal of Σ_A. We then perturb the final column of U_A by some angle Θ in order to form V_A, and the $\tilde{\epsilon}$-near-symmetric matrix A, with $\tilde{\epsilon} = A - A^{\mathrm{T}}$.

For our experiments, we construct a neural network that forward maps $X \mapsto AX$, for $X \in \mathbb{R}^{n\times s}$ a sketching matrix. Because we seek to learn a linear operator, we have opted to use linear activation functions. There are multiple reasons for this, firstly, because adding non-linearity to this map would make this network much slower at learning A. Although non-linear activation functions would still be able to understand the training data X we have provided, if we then started to apply unseen training data, the neural network would struggle reverse engineering the non-linearity in order to recover A.

Secondly, we had previously discussed that the neural network learning a matrix is an optimisation problem, since the network must append each bias and weight in order to minimise the loss function of the network. For linear activation functions, the loss is typically quadratic. Quadratic optimisation problems are much easier to solve in general, since they are convex, and hence have some useful properties that improve the convergence of gradient descent methods:

- Global convexity means the problem has one global minimum to converge to, meaning there are no local extrema that can capture gradient descent algorithms;

- Quadratic problems allow for rapid convergence towards the minimum point, or at least to the vicinity of the minimum.

We use this linear map to learn A by writing $\hat{A} = \hat{B}\hat{C}^{\mathrm{T}}$, then learning $\hat{A}X = \hat{B}\hat{C}^{\mathrm{T}}X = \hat{B}(\hat{C}^{\mathrm{T}}X)$, where $\hat{B}, \hat{C} \in \mathbb{R}^{100\times 10}$, where we have denoted $\hat{A}$ to be the neural network approximation of A.

We have assigned a budget of 80 matrix-vector queries, or a total of eight full rank preserving sketched X_i, for $1 \leq i \leq 8$. In keeping to the same approach as the rest of our research, we have opted for each of these matrices to have normally distributed entries. We have chosen each X_i to be Gaussian because of the high probability that such matrices are well-conditioned and invertible, as well as to not bias towards any particular subspace contained in $\mathbb{R}^n$. Furthermore, this means that $(X^{\mathrm{T}}AX)$ can be inverted, and so our findings from Theorem 2.3 and all preceding and succeeding remarks are relevant. Let $\Xi \coloneqq \{X_i : 1 \leq i \leq 8\}$ denote the set of queries. Then $\dim(\operatorname{span}(\Xi)) = \dim(\operatorname{span}(X_1, \ldots, X_8)) \leq 80$. In other words, we have only sampled up to 80 out of the 100-dimensional space. This means that $\dim(\operatorname{span}(\Xi^{\perp})) \geq 20$ and so our approximation $\hat{A}$ shall be unconstrained and A cannot be entirely recoverable. This is a trade-off we have decided upon since the options of increasing the budget, or biasing the net towards the structure of A would nullify our efforts hitherto. Increasing the budget closer to 100 matrix-vector queries would hinder our use of Gaussian matrices as inputs into the neural network, since for $s \simeq 100$ we would instead be best suited to trivially query A with the canonical basis vectors $\{e_j : 1 \leq j \leq 100\}$.

To speed up learning for gradient descent, we can try and form the best rank-r estimate of A before we try and it with the neural network. Let $Y_i = AX_i$, for each i. Then, we shall denote

$$X_C = [X_1 \cdots X_8] \text{ and } Y_C = [Y_1 \cdots Y_8], \text{ for } X_C, Y_C \in \mathbb{R}^{100\times 80} \tag{2.59}$$

to be the concatenations of the inputs and outputs, respectively. Now, we shall consider the least squares solution that maps onto $\operatorname{span}(X_C)$, denoted A_{ls}

$$A_{\mathrm{ls}} \coloneqq \arg\min_{A} \|AX_C - Y_C\|_F^2. \tag{2.60}$$

This is given by $A_{\mathrm{ls}} = Y_C X_C^{\dagger} + W(I - X_C X_C^{\dagger})$ in general for any arbitrary matrix $W \in \mathbb{R}^{100\times 100}$. In particular, we find the minimiser in the Frobenius norm when $W = 0$. This is an exact interpolant since $A_{\mathrm{ls}}X_C = Y_C X_C^{\dagger} X_C = AX_C X_C^{\dagger} X_C = AX_C$ by the properties of the pseudoinverse. Since we have assumed that the rank of A is known, we shall then truncate A_{ls} to rank-r using its SVD. Writing $A_{\mathrm{ls}} = U_{\mathrm{ls}}\Sigma_{\mathrm{ls}}V_{\mathrm{ls}}^{\mathrm{T}}$, it is well-known that the best low rank approximation of any matrix is given by its truncated SVD:

$$A_{\mathrm{ls},r} = \sum_{k=1}^{r} \sigma_{\mathrm{ls},j} u_{\mathrm{ls},j} v_{\mathrm{ls},j}^{\mathrm{T}}, \tag{2.61}$$

where $u_{\mathrm{ls},j}, \sigma_{\mathrm{ls},j}, v_{\mathrm{ls},j}$ represent the columns of U_{ls}, Σ_{ls}, and V_{ls} respectively. To ensure the numerical stability of this matrix, we shall then write

$$\hat{B}_{\mathrm{ls},r} \coloneqq U_{\mathrm{ls}[:,1:r]} \Sigma^{\frac{1}{2}}_{\mathrm{ls}[1:r,1:r]} \text{ and } \hat{C}_{\mathrm{ls},r} \coloneqq V_{\mathrm{ls}[:,1:r]} \Sigma^{\frac{1}{2}}_{\mathrm{ls}[1:r,1:r]}, \tag{2.62}$$

where we have used MATLAB notation to denote the truncated sections of the original matrices, U_{ls}, Σ_{ls}, and V_{ls}.

For the training on the matrix, we have used the relative mean squared error in the Frobenius norm as the loss function, denoted $\mathcal{L}$. For each input X_i and output AX_i

$$\mathcal{L}_i = \frac{\|\hat{B}(\hat{C}^{\mathrm{T}}X_i) - AX_i\|_F^2}{\|AX_i\|_F^2}. \tag{2.63}$$

We also add an additional loss of regularisation of the matrices $\hat{B}$ and $\hat{C}$, given by $\mathcal{L}_{\mathrm{reg}}$:

$$\mathcal{L}_{\mathrm{reg}} \coloneqq \|\hat{B}^{\mathrm{T}}\hat{B} - \hat{C}^{\mathrm{T}}\hat{C}\|_F^2. \tag{2.64}$$

We add this extra term to penalise cases where $\hat{B}$ and $\hat{C}$ are of greatly differing orders of magnitude. In this case, the gradient descent methods, such as backpropagation, would scale with the size of the matrices, and hence one matrix would dominate the adjustment of their respective biases. We scale this error by a constant, μ, depending n how much we wish to penalise this relative to $\mathcal{L}$, the loss function for the data. If μ is too small, there would not be sufficient punishment for numerical instabilities in the calculation of the gradients for learning, and if μ was too large, our network would favour keeping both matrices as similar to each other in magnitude over the learning itself. In total, we have

$$\mathcal{L}_{\mathrm{total}} = \sum_i \mathcal{L}_i + \mu\mathcal{L}_{\mathrm{reg}}, \tag{2.65}$$

and in our experiments we had taken $\mu = 1 \times 10^{-3}$.

We have used the PyTorch Python library to build our neural network, and use standard backpropagation available through `.backward()` in order to calculate the loss gradients $\frac{\partial\mathcal{L}_{\mathrm{total}}}{\partial\hat{B}}$ and $\frac{\partial\mathcal{L}_{\mathrm{total}}}{\partial\hat{C}}$. We then use `Adam` in order to optimise each parameter, using the algorithm stated in Figure 2.8. We had set the learning rate given by γ, to be the step size that is used in order to minimise the loss function of the neural network. We start initially with $\gamma = 3\times10^{-3}$, though we have used `ReduceLROnPlateau(optimizer, patience=100, factor=0.5)` in our code when the loss function plateaus in order to speed up convergence. Weight decay, λ, is the parameter used for the regularisation of the loss function during training. This is to prevent over-fitting or under-fitting of learned data, which would inhibit the network to learn unseen data in both cases. For our experiments we had taken $\lambda = 1 \times 10^{-6}$. The constants β_1 and β_2 are known as exponential moving averages, are used to understand how the progression of weights change through each iteration. As the neural network learns better weights, we should expect those of previous iterations to be less important. PyTorch defaults these to $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and this is what we have used. We have used ε to denote a small parameter to ensure that there is no division by zero. We have taken $\varepsilon = 1\times10^{-8}$, and this should not be confused with the near symmetry we have used in previous sections, ϵ and $\tilde{\epsilon}$. We have used ϑ in order to denote the trainable parameters, in our case, we have that $\vartheta = \{\hat{B}, \hat{C} \in \mathbb{R}^{n\times r} : \hat{A} = \hat{B}\hat{C}^{\mathrm{T}} \simeq A\}$. For every step i, g_i is the gradient of the loss function, and m_i and v_i represent the exponential moving averages of the gradient and gradient squared, respectively. We leave both `maximise` and `amsgrad` set to `False`. In our case, $i = 1, \ldots, 8$, the number of input matrices, and for each step we have assigned 1500 epochs.

Figure 2.9 shows us the results of our experiment in matrix recovery of the neural network against the the Nyström approximation of a near-symmetric matrix. The procedure for the Nyström approximation remains as before, by forming a near-symmetric matrix by perturbing the right singular vector or vectors of a symmetric matrix. In this case, we had perturbed the singular vector corresponding to the smallest singular value, $\sigma_r = 1 \times 10^{-3}$. Of the possible inputs $\{X_1, \ldots, X_8\}$, we had chosen the most well-conditioned for the Nyström method, in order to minimise the numerical uncertainties in the inversion of the kernel matrix. We had decided upon using the relative errors for each method, therefore, the orange line in Figure 2.9 represents the expression $\frac{\|\tilde{A}-A\|_F}{\|A\|_F}$, and for the neural network we have $\frac{\|\hat{A}-A\|_F}{\|A\|_F}$. We see in Figure 2.10 that the relative near symmetry error, given by $\frac{\|\tilde{\epsilon}\|_F}{\|A\|_F}$ is a good match with the Nyström approximation. This is to be expected because we had chosen a situation that guarantees a good agreement between both through Theorem 2.3. The problem faced by the neural network, is that its learning is only constrained to $\mathrm{span}(\Xi)$, so for

Algorithm 1 Adam PyTorch Optimiser

Require: γ (learning rate), β_1, β_2 (betas), ϑ_0 (parameters), $\mathcal{L}_{\text{total}}(\vartheta)$ (objective function), λ (weight decay), `amsgrad`, `maximize`, ε (epsilon)

Initialize $m_0 \leftarrow 0$ (first moment), $v_0 \leftarrow 0$ (second moment), $v_0^{\max} \leftarrow 0$
for $i = 1, 2, \ldots$ **do**
  if `maximize` **then**
    $g_i \leftarrow -\nabla_\vartheta \mathcal{L}_{\text{total},i}(\vartheta_{i-1})$
  else
    $g_i \leftarrow \nabla_\vartheta \mathcal{L}_{\text{total},i}(\vartheta_{i-1})$
  end if
  if $\lambda \neq 0$ **then**
    $g_i \leftarrow g_i + \lambda \vartheta_{i-1}$
  end if
  $m_i \leftarrow \beta_1 m_{i-1} + (1 - \beta_1) g_i$
  $v_i \leftarrow \beta_2 v_{i-1} + (1 - \beta_2) g_i^2$
  $\hat{m}_i \leftarrow m_i / (1 - \beta_1^i)$
  if `amsgrad` **then**
    $v_i^{\max} \leftarrow \max(v_{i-1}^{\max}, v_i)$
    $\hat{v}_i \leftarrow v_i^{\max} / (1 - \beta_2^i)$
  else
    $\hat{v}_i \leftarrow v_t / (1 - \beta_2^i)$
  end if
  $\vartheta_i \leftarrow \vartheta_{i-1} - \gamma \hat{m}_i / (\sqrt{\hat{v}_i} + \varepsilon)$
end for
return ϑ_i

Figure 2.8: The pseudocode for the Adam stochastic optimisation algorithm, the code for which is available in [13] and is based on the work of Kingma and Ba [11].

any $\hat{A}$ and any vector $x \in \mathbb{R}^n$, we can only learn that $\hat{A}x = Ax$, for all $x \in \text{span}(\Xi)$, and we have no method of minimising the error of the part of A that lies in $\text{span}(\Xi^\perp)$. We shall now show why this results in an error of $\mathcal{O}(10^{-1})$. Let us define $X_{\text{new}} \in \mathbb{R}^{n \times r}$ to be a new Gaussian matrix. Denote P_Ξ be the or onto $\text{span}(\Xi)$ and let $P_{\Xi^\perp}$ be the

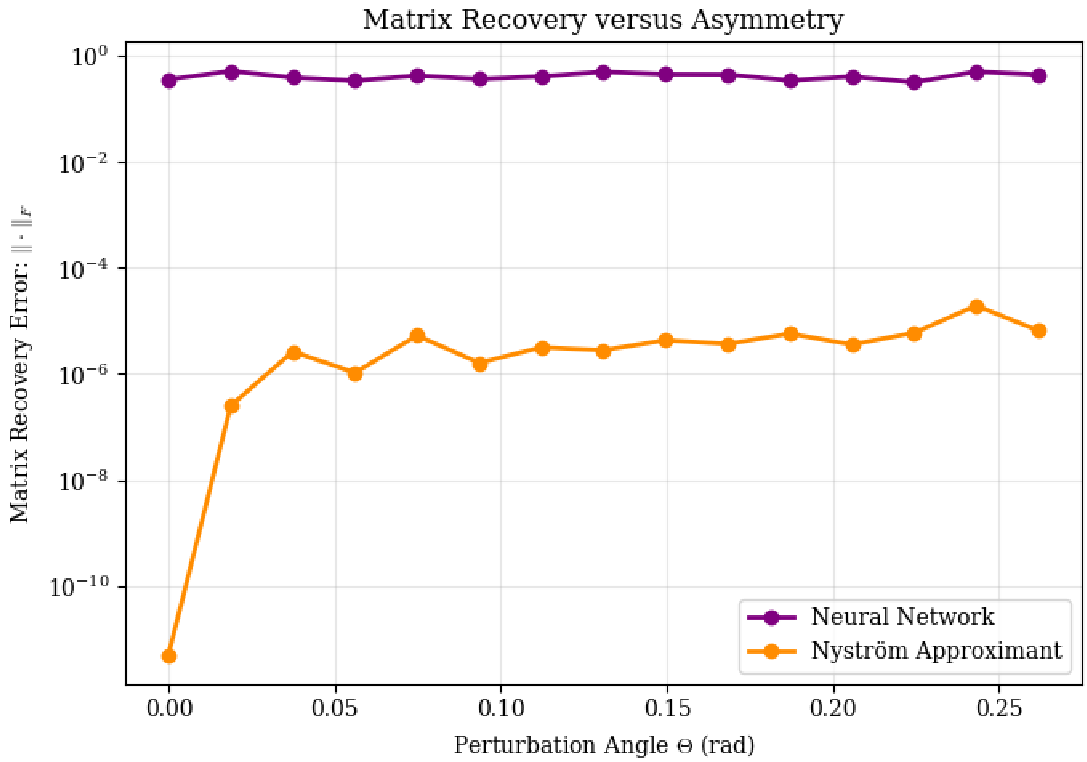


Figure 2.9: A plot showing the relative matrix recovery error in the Frobenius norm of the neural network described above and the Nyström approximation, against perturbation angle Θ of the right singular vector corresponding to the smallest singular value.

complementary projector onto $\operatorname{span}(\Xi^{\perp})$. Then, we can decompose X_{new} into

$$X_{\text{new}} = P_{\Xi} X_{\text{new}} + P_{\Xi^{\perp}} X_{\text{new}}, \tag{2.66}$$

and hence it follows that $AX_{\text{new}} = AP_{\Xi}X_{\text{new}} + AP_{\Xi^{\perp}}X_{\text{new}}$ Since the neural network can learn the action of A onto $\operatorname{span}(\Xi)$ to high precision, but cannot infer anything from its action upon $\operatorname{span}(\Xi^{\perp})$, we have that

$$\|\hat{A}X_{\text{new}} - AX_{\text{new}}\|_F^2 \geq \|AP_{\Xi^{\perp}}X_{\text{new}}\|_F^2. \tag{2.67}$$

The projector $X_{\text{new}}X_{\text{new}}^{\mathrm{T}}$ follows a Wishart distribution (see Anderson [1]), that we shall denote $X_{\text{new}}X_{\text{new}}^{\mathrm{T}} \sim \mathcal{W}_n(I_r, r)$, for I_r the scale matrix and r the degrees of freedom. Then, we find the standard result

$$\mathbb{E}\left[X_{\text{new}}X_{\text{new}}^{\mathrm{T}}\right] = rI_n, \tag{2.68}$$

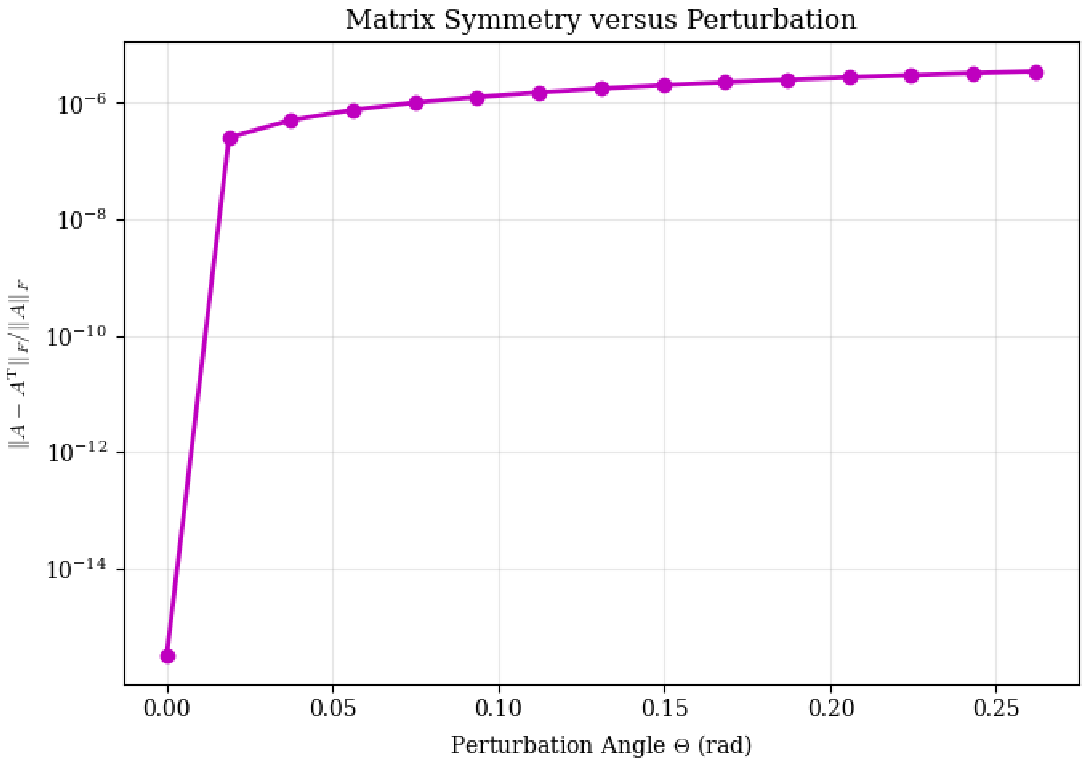


Figure 2.10: A plot to show the relative symmetry error in the Frobenius norm of the $\tilde{\epsilon}$-near-symmetric matrix A as a function of perturbation angle Θ. We see a good correlation between this function and the relative error size of the Nyström approximant in Figure 2.9

and since for any vector $\mathsf{x} \sim \mathcal{N}(\mathsf{m}, \mathsf{s})$, we have that $\mathbb{E}\left[\mathsf{x}^{\mathrm{T}} A \mathsf{x}\right] = \mathrm{tr}(A\mathsf{s}) + (\mathsf{m}^{\mathrm{T}} A \mathsf{m})$. It follows that since $\mathbb{E}\left[X_{\mathrm{new}}^{\mathrm{T}} A X_{\mathrm{new}}\right] = \mathbb{E}\left[\|AX_{\mathrm{new}}\|_F^2\right]$, we have that

$$\mathbb{E}\left[\|AX_{\mathrm{new}}\|_F^2\right] = r\,\mathrm{tr}(AA^{\mathrm{T}}). \tag{2.69}$$

Then, if X_{new} is Gaussian, then $P_{\Xi^\perp}$ is a Gaussian projection onto $\mathrm{span}(\Xi^\perp)$. As a result,

$$\mathbb{E}\left[\|AP_{\Xi^\perp} X_{\mathrm{new}}\|_F^2\right] = r\,\mathrm{tr}(P_{\Xi^\perp} AA^{\mathrm{T}}). \tag{2.70}$$

Since P_Ξ is Gaussian but AA^{T} is entirely deterministic, we have that $\mathbb{E}\left[\mathrm{tr}(\mathrm{P}_\Xi \mathrm{AA}^{\mathrm{T}})\right] = \mathrm{tr}\left(\mathbb{E}\left[P_\Xi\right] AA^{\mathrm{T}}\right)$. The projector P_Ξ should have an expectation value equal to $\mathbb{E}\left[P_\Xi\right] = \frac{\dim(\Xi)}{n} I_n$, since $\mathrm{tr}(P_\Xi) = \mathrm{rank}(P_\Xi) = \dim(\Xi)$, and so $\mathrm{tr}\left(\mathbb{E}\left[P_\Xi\right] AA^{\mathrm{T}}\right) = \frac{\dim(\Xi)}{n}\,\mathrm{tr}\left(AA^{\mathrm{T}}\right)$. Combining this with Equation (2.69), we have that

$$\mathbb{E}\left[\frac{\|AP_{\Xi^\perp} X_{\mathrm{new}}\|_F^2}{\|AX_{\mathrm{new}}\|_F^2}\right] = 1 - \frac{\dim(\Xi)}{n}. \tag{2.71}$$

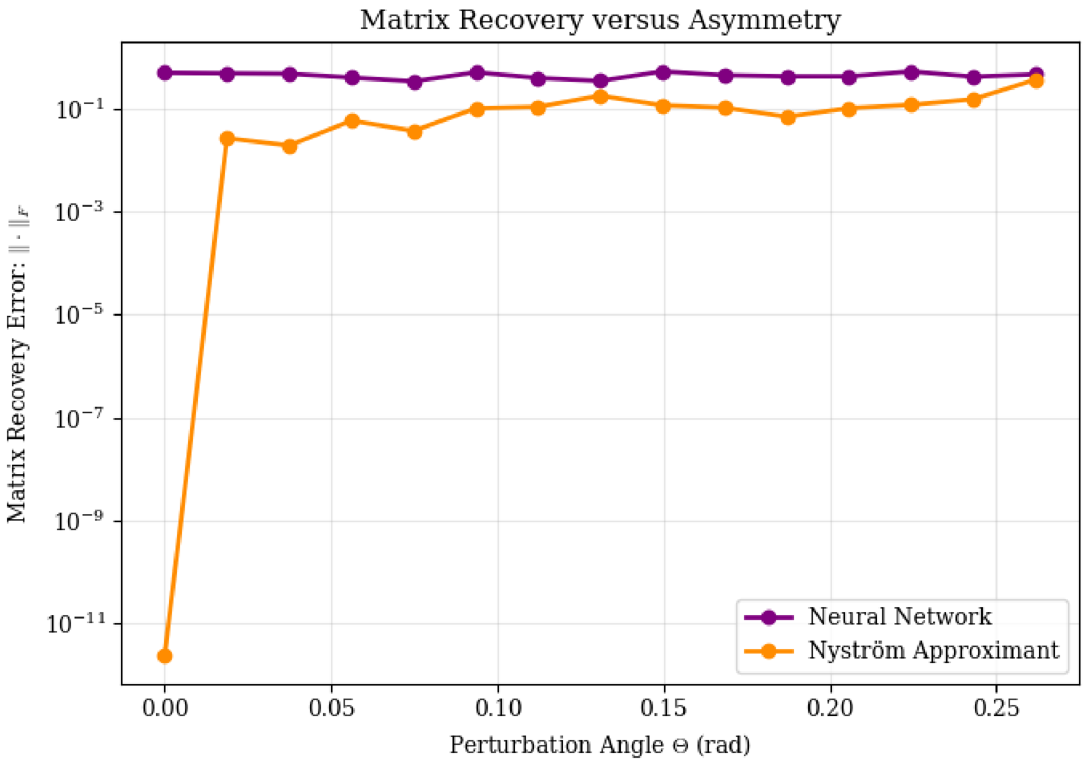


Figure 2.11: A plot of the error of neural network and Nyström methods in the Frobenius norm for matrix recovery as a function of perturbation angle Θ, though in this case, we have perturbed the right singular vector corresponding to the largest singular value.

In our case, we have $\dim(\Xi) = 80$ and $n = 100$, and hence we find our expectation error to be around 0.2, or $\mathcal{O}(10^{-1})$, as we had expected. We see in Figure 2.11 that the errors of both methods are now of similar magnitude. This is because we have now perturbed the singular vector corresponding to the largest singular value, and in this case, we find that the relative errors of both methods coincide. From Equation (2.71), we see that the error can be reduced by learning $\operatorname{span}(\Xi^{\perp})$.

Finally, we shall now seek to estimate the size of $\frac{\|\tilde{A}-A\|_F}{\|A\|_F}$, in order to demonstrate the matrix recovery error for the Nyström approximation flattens around $\mathcal{O}(10^{-5})$ to $\mathcal{O}(10^{-4})$. Let us return to expression (2.2). Then, by writing $A = U_A \Sigma_A V_A^{\mathrm{T}}$ to be the thin SVD of A, and for any X a Gaussian matrix, let us denote $G_{U_A} := U_A^{\mathrm{T}} X \in \mathbb{R}^{r\times r}$ and $G_{V_A} := V_A^{\mathrm{T}} X \in \mathbb{R}^{r\times r}$ to be Gaussian matrices, and let us assume that they are invertible. It follows that

$$AX(X^{\mathrm{T}}AX)^{-1} = U_A \Sigma_A G_{V_A} \left(G_{U_A}^{\mathrm{T}} \Sigma_A G_{V_A} \right)^{-1} = U_A G_{U_A}^{-\mathrm{T}}. \tag{2.72}$$

By combining the above expression with Equation (2.2), we have that

$$\|\tilde{A}X - AX\|_F \leq \|G_{U_A}^{-1}\|_2 \|X^{\mathrm{T}}\tilde{\epsilon}X\|_F. \tag{2.73}$$

From the Wishart distribution, we have that $\mathbb{E}[\|AX\|_F] = r\|A\|_F^2$, and that for the $(i,j)^{\text{th}}$ entries of $(X^{\mathrm{T}}\tilde{\epsilon}X)$, denoted $(X^{\mathrm{T}}\tilde{\epsilon}X)_{ij}$, we have $\mathbb{E}[\|(X^{\mathrm{T}}\tilde{\epsilon}X)_{ij}\|_F^2] = \mathrm{tr}(\tilde{\epsilon}\tilde{\epsilon}^{\mathrm{T}}) = \|\tilde{\epsilon}\|_F^2$ by the previous relation. Then summing through each i and j in $1, \ldots, r$, we have

$$\sum_{\substack{i,j=1 \\ i \neq j}}^{r} \mathbb{E}[(X^{\mathrm{T}}\tilde{\epsilon}X)_{ij}^2] = r(r-1)\|\tilde{\epsilon}\|_F^2. \tag{2.74}$$

As a result, we can estimate $\|X^{\mathrm{T}}\tilde{\epsilon}X\|_F \simeq \sqrt{r(r-1)}\|\tilde{\epsilon}\|_F$. For spectral norm, we have $\|G_{U_A}^{-1}\|_2 \simeq \frac{1}{\sqrt{r}}$ with high probability, and hence we find that

$$\frac{\|\tilde{A}X - AX\|_F}{\|AX\|_F} \lesssim \frac{\sqrt{r(r-1)}}{r} \frac{\|\tilde{\epsilon}\|_F}{\|A\|_F}. \tag{2.75}$$

For matrices of much larger dimension and rank, $\sqrt{r(r-1)} \simeq r$, in which case the error of the Nyström approximation is dominated by the relative sizes of the $\tilde{\epsilon}$-near-symmetry compared to the matrix A itself.

Chapter 3

Conclusions and Future Work

To conclude, we had began this dissertation by introducing operator learning and its importance in the present day. We had then focused on learning linear operators such as matrices, and highlighted the tension between seeking to learn as much of a matrix as possible in as few queries as possible. This lead us to present two different methods for matrix learning, namely dynamic mode decomposition and the Nyström approximation. Imposing another restriction on our research by attempting to learn linear operators without access to their adjoint, this lead us naturally to the study of symmetric matrices. To this end, we had then considered the benefits of the Nyström approximation, namely that it could recreate a symmetric positive semi-definite matrix to machine precision under the right circumstances, and we had conjectured that such a method can be used to learn near symmetric matrices up to an error governed chiefly by the amount of asymmetry displayed in the matrix.

Next, we had introduced a definition of ϵ-near-symmetry, as defined in Boullé, Halikias, *et al.* [2]. We had made some progress in exploring the properties of ϵ-near-symmetric matrices, by considering the left and right singular vectors of these matrices and concluding that this definition of near-symmetry is principally a relation of the canonical angle Θ between the left and right singular vectors of the matrix. We had also considered how matrices with the same value of ϵ-near-symmetry cannot be trivially recreated with the Nyström approximation to equal error because of the inherent importance of the energy of the perturbed singular subspace, as well as the angle of perturbation.

This lead us in the next chapter to define $\tilde{\epsilon}$-near-symmetry, which would seek to incorporate both the perturbation angles between the singular vectors of a near symmetric matrix, as well as the magnitudes of the singular values that correspond to them. We had used this notion of near-symmetry in order to provide a probabilistic theorem to bound the operator error between the $\tilde{\epsilon}$-near-symmetric matrix and its Nyström approximant as a function of its dimension, sketching matrix and asymmetry.

In our numerical experiments, we had tested the sharpness of this upper bound. The analysis of this section was challenging because of the lack of research into this topic, but we had sought to tighten this bound by removing as many norm inequalities as possible. This lead us to the empirical discovery of the use of inverted Gaussian matrices to supplement our theorem which had lead to a closer approximation of the error bound in some cases.

Following this, we would analyse the notion of $\tilde{\epsilon}$-near-symmetry. We found that we can also relate the norm of $\tilde{\epsilon}$ to the same dependencies on the perturbation angles between the left and right singular vectors of a matrix as in the case of ϵ-near-symmetric matrices, with the added benefit that our definition of asymmetry also considers the energy of the perturbed components.

Lastly, we had created a neural network which would seek to learn a near-symmetric matrix from an input sketch and the output sketched matrix. We found, as with dynamic mode decomposition, that the error in the recreation of the matrix is governed by the dimension of the sketch, and hence for small sketches and near symmetric matrices, the Nyström approximation is better suited.

Without time constraints, there are two sections of our analysis we would wish to work on further in the future. Ideally we would have liked to spend more time attempting to sharpen the upper bound stated in Theorem 2.3. Due to the lack of literature on our studies, it is difficult to gauge the tightness of this bound without more experiments of our own. Secondly, we would wish to explore further the use of Gaussian matrices such as in Equation (2.34), where the data showed promising results on the approximation of the magnitude of $\|A - \tilde{A}\|$. Again, we have not seen such methods employed elsewhere, and if time permitted, we would have explored the mathematical reasons for their use, rather than only presenting the empirical results of our numerical experiments.

Finally, we hope that our research on the learning of near-symmetric matrices is extended further, with the notions of $\tilde{\epsilon}$-near symmetry, the upper bound provided by Theorem 2.3 and studies into the approximation of $\|A - \tilde{A}\|$ by random matrices serving as a basis of that work.

References

[1] T. W. Anderson. *An Introduction to Multivariate Statistical Analysis*. Wiley, 2003.

[2] N. Boullé, D. Halikias, *et al.* “Operator Learning without the Adjoint”. In: *Journal of Machine Learning Research* (2024).

[3] C. Davis and W. M. Kahan. “The Rotation of Eigenvectors by a Perturbation”. In: *Society for Industrial and Applied Mathematics Journal on Numerical Analysis* (1970).

[4] P. Drineas and M. W. Mahoney. “On the Nyström Method for Approximating a Gram Matrix for Improved Kernel-Based Learning”. In: *Journal of Machine Learning Research* (2005).

[5] K. Fan and A. J. Hoffman. “Some Metric Inequalities in the Space of Matrices”. In: *Proceedings of the American Mathematical Society* (1955).

[6] D. Halikias and A. Townsend. “Structured Matrix Recovery from Matrix-Vector Products”. In: *Numerical Linear Algebra with Applications* (2023).

[7] C. R. Harris, K. J. Millman, *et al.* “Array programming with NumPy”. In: *Nature* 585.7825 (2020), pp. 357–362. doi: 10.1038/s41586-020-2649-2. url: https://doi.org/10.1038/s41586-020-2649-2.

[8] K. Hornik, M. Stinchcombe, *et al.* “Universal Approximation of an Unknown Mapping and its Derivatives Using Multilayer Feedforward Networks”. In: *Neural Networks* (1990).

[9] H.-Y. Huang, R. Kueng, *et al.* “Provably efficient machine learning for quantum many-body problems”. In: *Science* (2022).

[10] J. Jumper, R. Evans, *et al.* “Highly accurate protein structure prediction with AlphaFold”. In: *Nature* (2021).

[11] D. P. Kingma and J. Ba. “Adam: A Method for Stochastic Optimization”. In: *3rd International Conference on Learning Representations (ICLR)*. 2015. url: `https://arxiv.org/abs/1412.6980`.

[12] W. S. McCulloch and W. Pitts. “A Logical Calculus of the Ideas Immanent in Nervous Activity”. In: *Bulletin of Mathematical Biophysics* (1943).

[13] PyTorch Contributors. *torch.optim.Adam — PyTorch Documentation*. Implementation based on [11]. 2024. url: `https://pytorch.org/docs/stable/generated/torch.optim.Adam.html` (visited on 08/29/2024).

[14] M. Rudelson. “Invertibility of Random Matrices: the Norm of the Inverse”. In: *Annals of Mathematics* (2008).

[15] O. Russakovsky, J. Deng, *et al*. “ImageNet Large Scale Visual Recognition Challenge”. In: *International Journal of Computer Vision* (2015).

[16] G. W. Stewart. “On the Perturbation of Pseudo-Inverses, Projections and Linear Least Squares Problems”. In: *Society for Industrial and Applied Mathematics* (1977).

[17] S. J. Szarek. “Condition Numbers of Random Matrices”. In: *Journal of Complexity* (1991).

[18] J. H. Tu. “Dynamic Mode Decomposition: Theory and Applications”. PhD thesis. Princeton University, 2013.

[19] P. Virtanen, R. Gommers, *et al*. “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python”. In: *Nature Methods* 17 (2020), pp. 261–272. doi: `10.1038/s41592-019-0686-2`.